

063-0605-00L : Computational Structural Design 1

Computational Graphic Statics

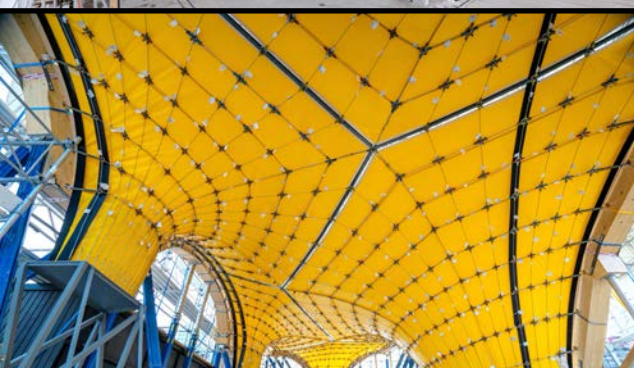
Dr. Juney Lee & Dr. Lluís Enríque

Autumn Semester 2021



BRG
<http://block.arch.ethz.ch>

 @blockresearchgroup



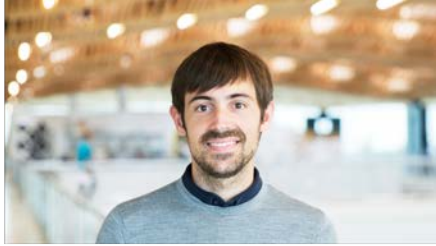


BRG

Lecturers



Dr. Juney Lee



Dr. Lluís Enrique

PhD student instructors



Lotte Aldinger



Ricardo Avelino

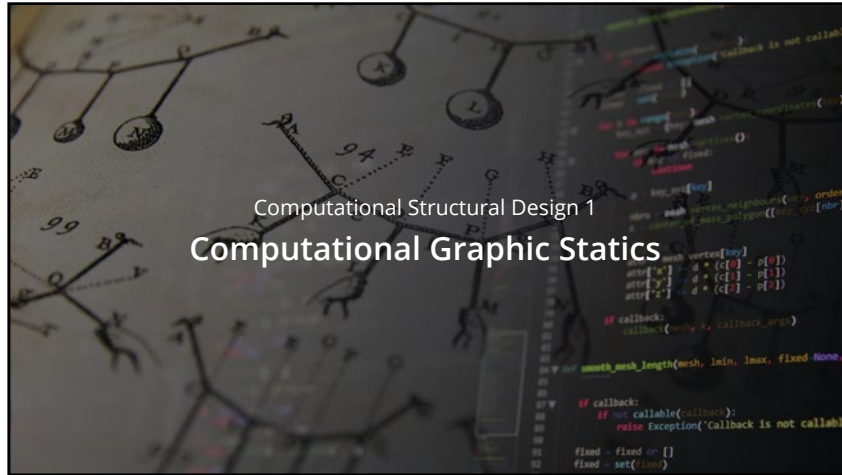
Teaching assistant



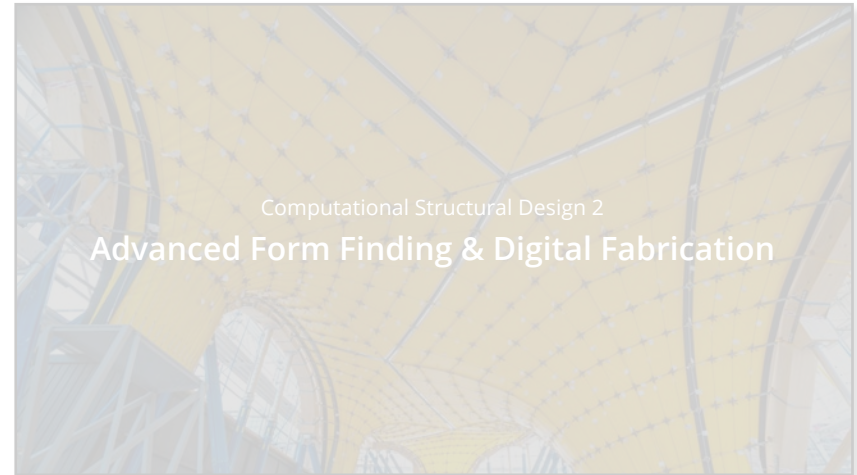
Kerstin Spiekermann



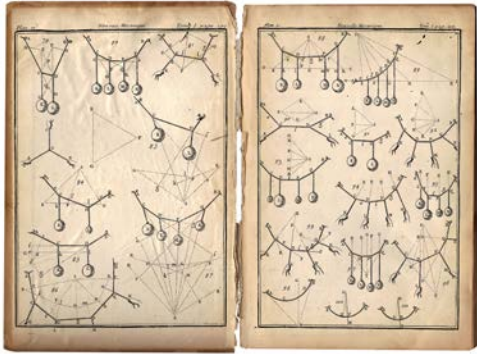
Michele Capelli



Fall 2021



Spring 2022



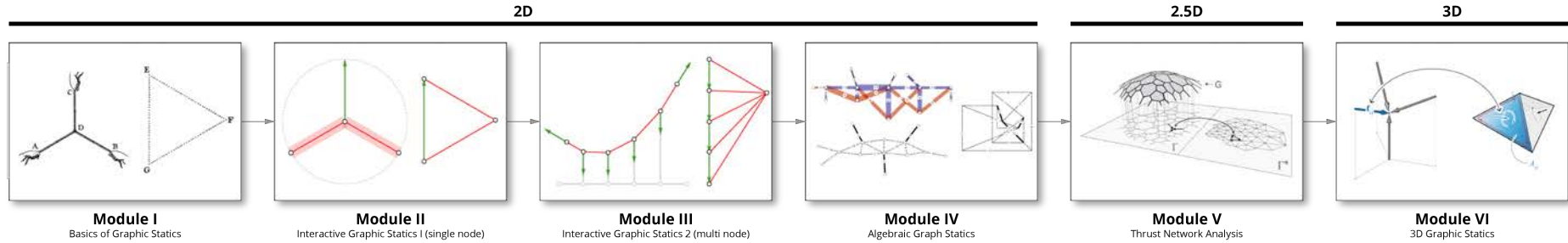
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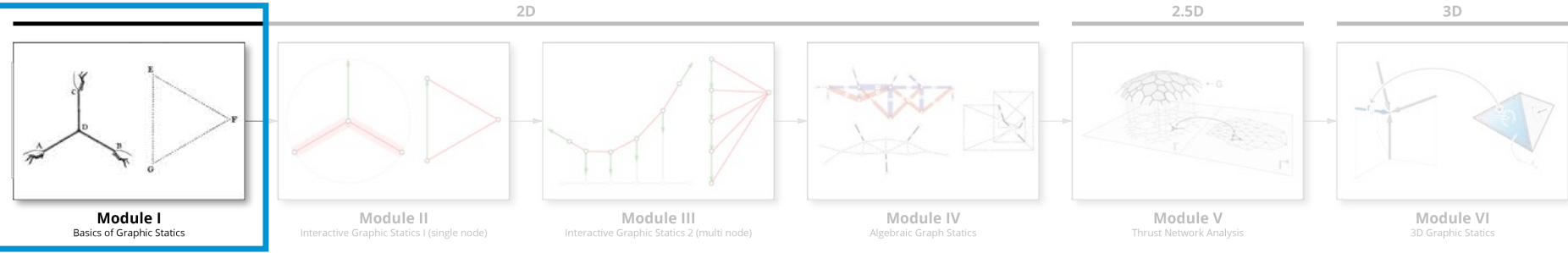


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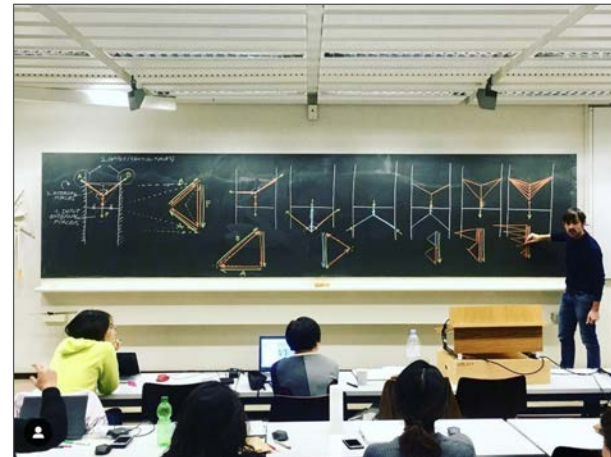
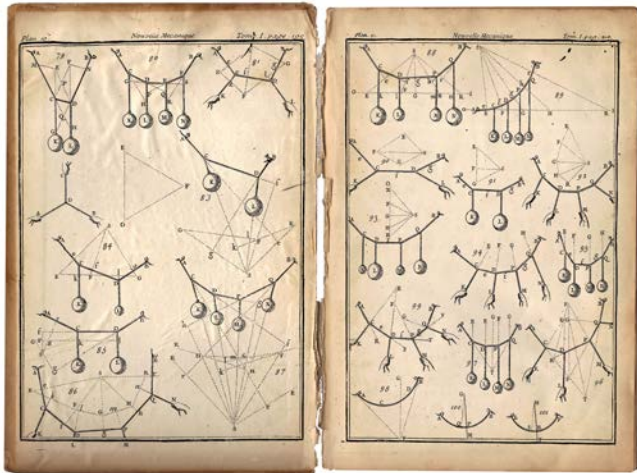


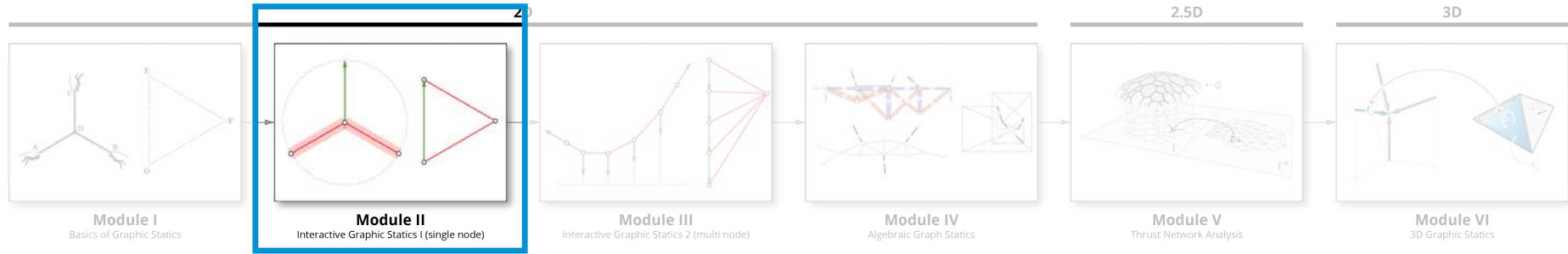
Explore new **structural design opportunities**
by combining **graphic statics** with **computational tools**



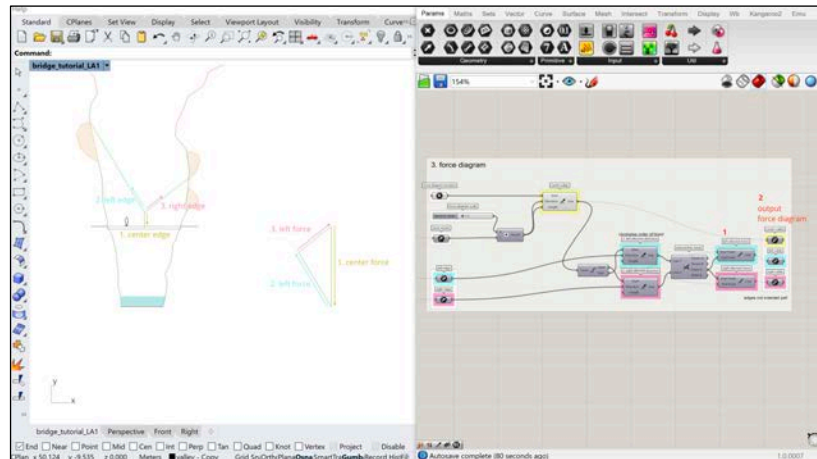


Module I · Basics of graphic statics



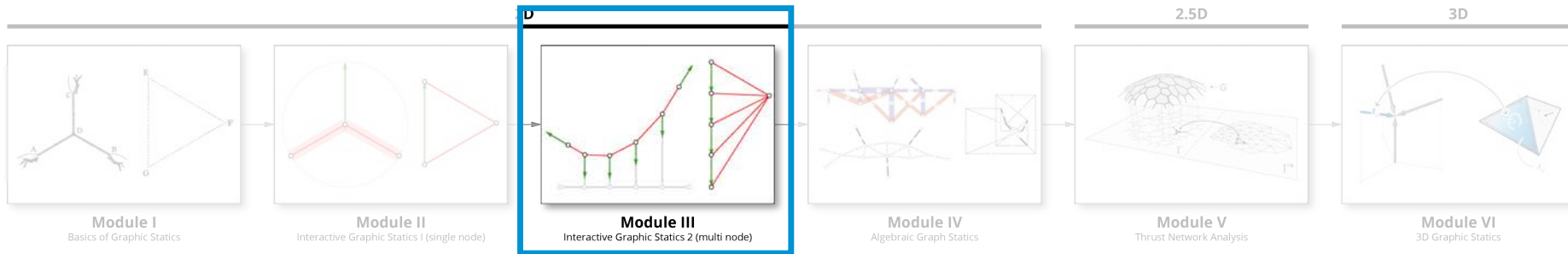


Module II · 2D procedural graphic statics 1

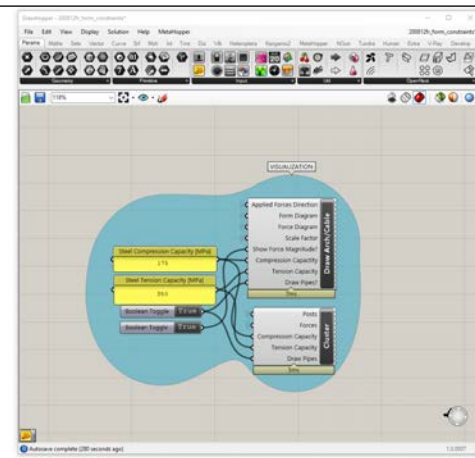
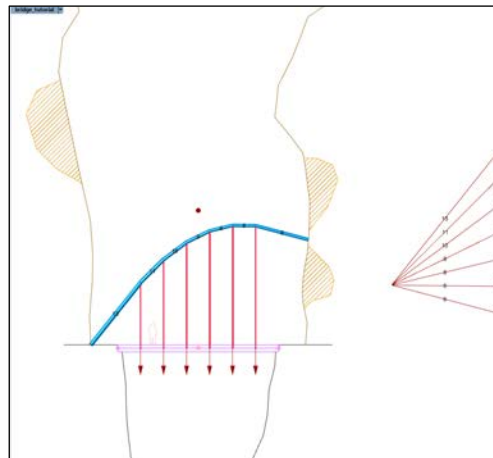
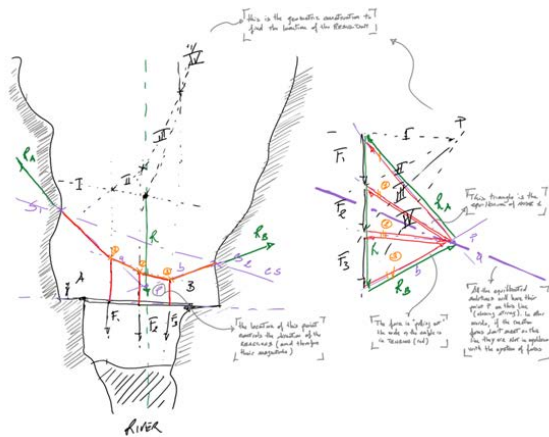


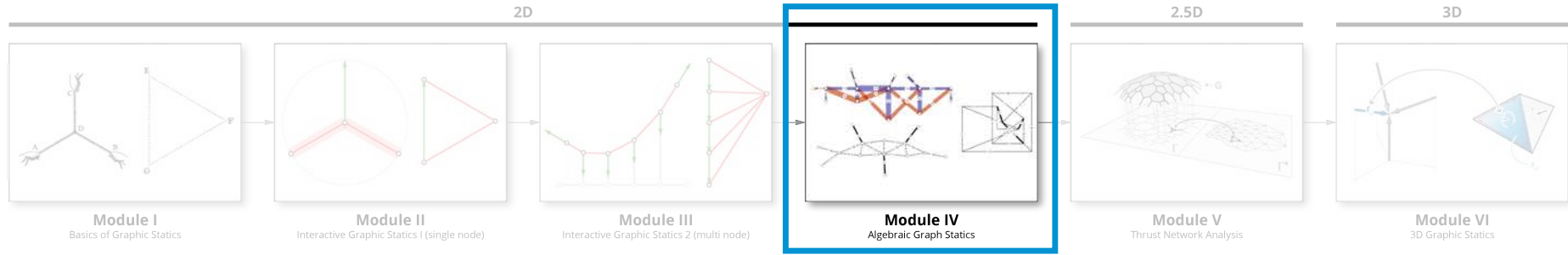
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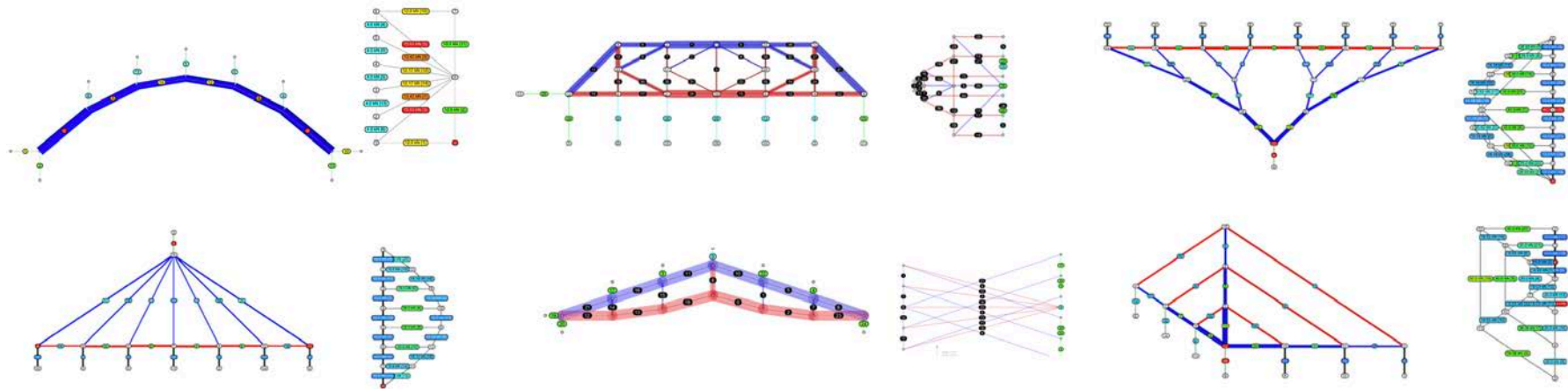


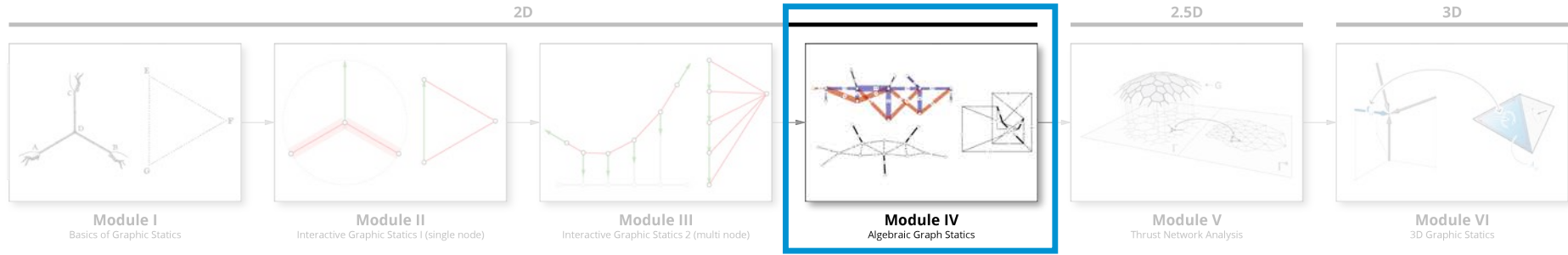
Module III · 2D procedural graphic statics 2





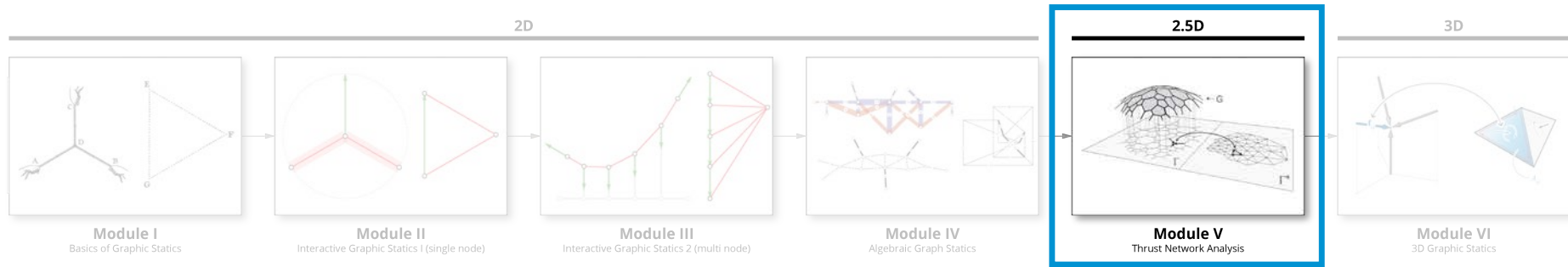
Module IV · Algebraic graph statics (automatic construction)



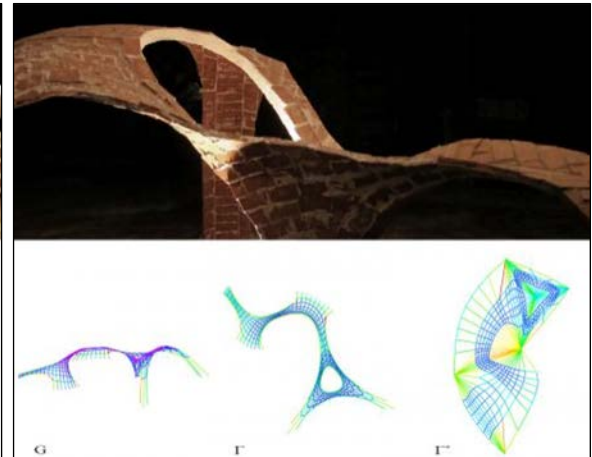
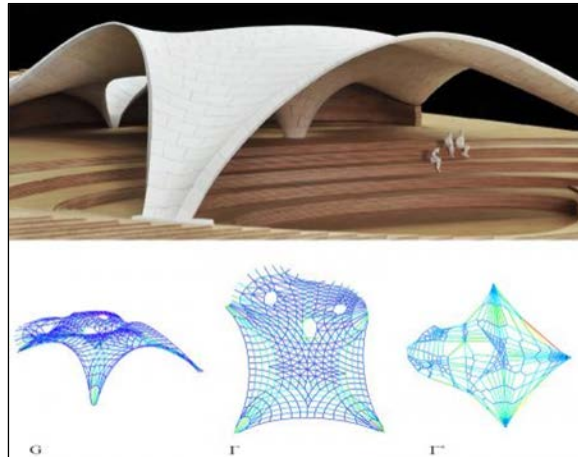
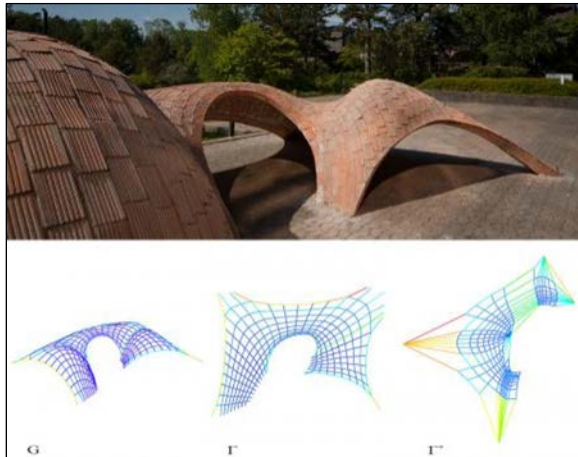


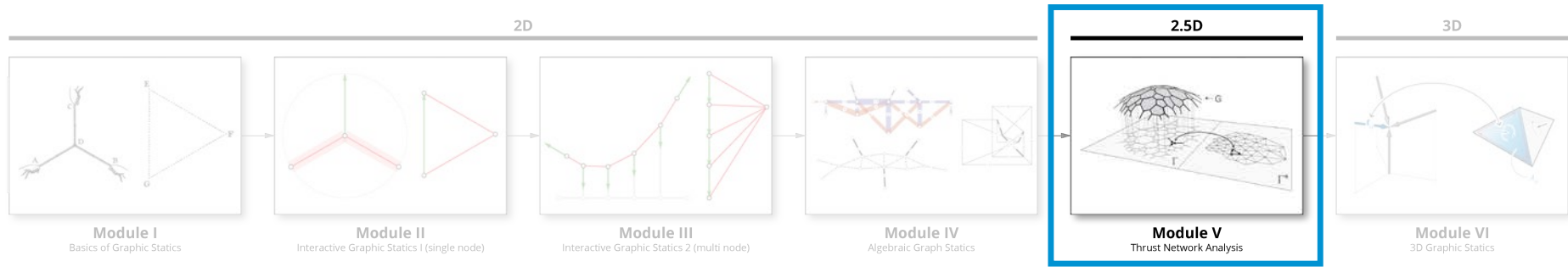
Module IV · Algebraic graph statics (automatic construction)



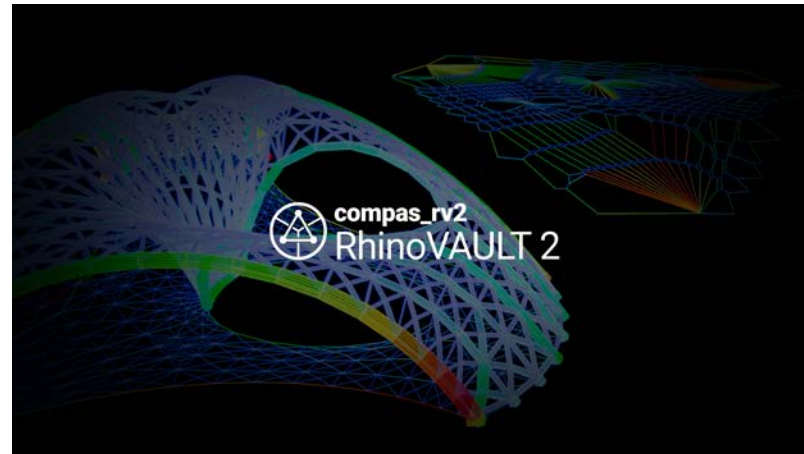


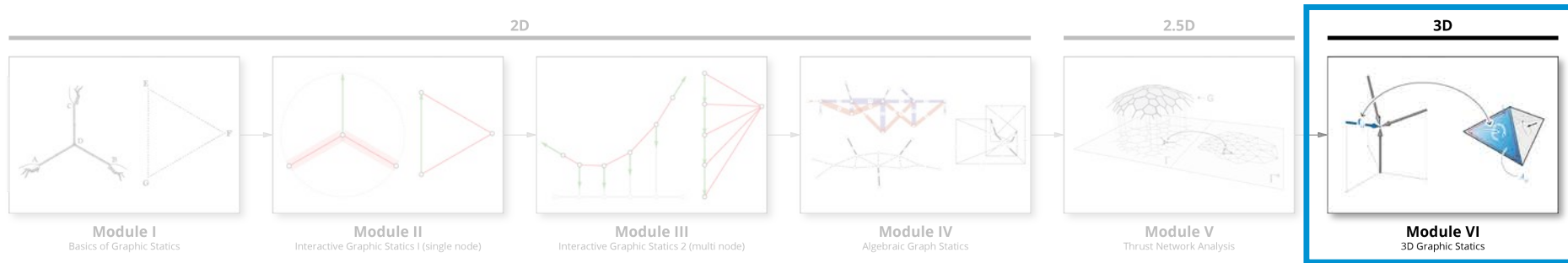
Module V · Thrust Network Analysis & RhinoVAULT



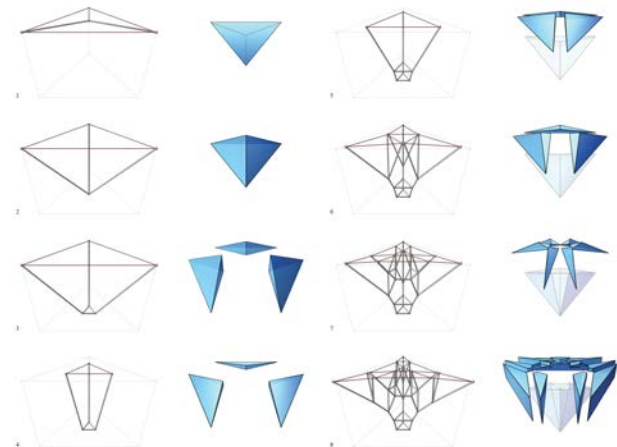
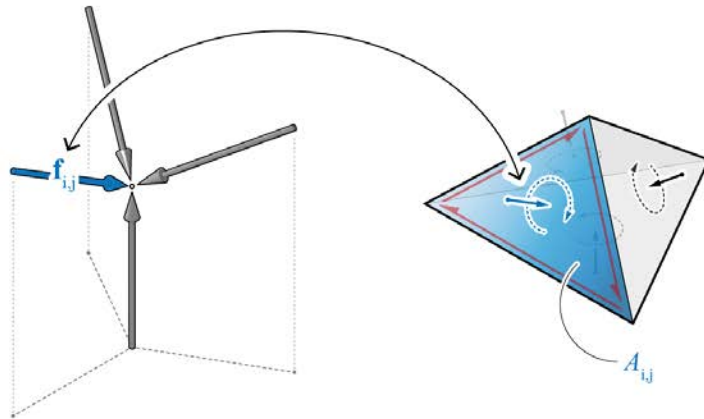


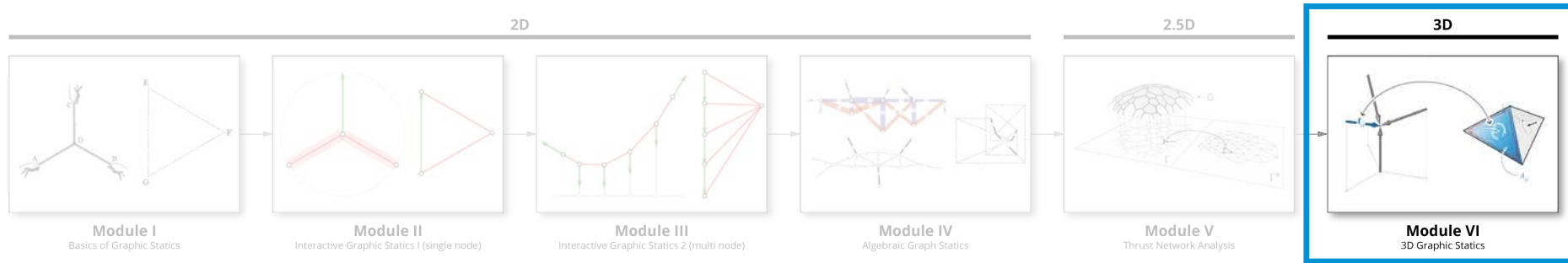
Module V · Thrust Network Analysis & RhinoVAULT





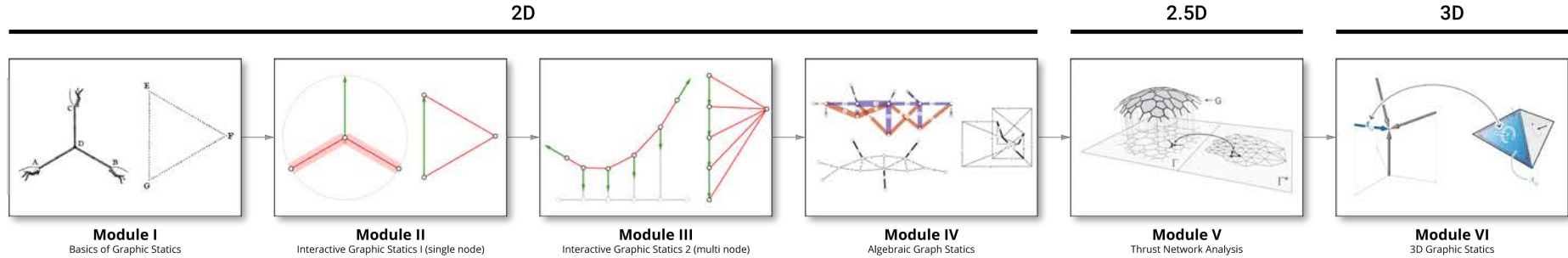
Module VI · 3D graphic statics





Module VI · 3D graphic statics





Structural Typology	Topic	Module	Week	hr	Topic	Instructor		
1. planar structures	cables & arches	I	Week 1 (9/24)	1	Lecture 1 Introduction to graphic statics	Dr. Juney Lee		
				2	Tutorial 1 Basics of graphic statics on "blackboard" Introduction to Rhino + Grasshopper (GH)	Dr. Lluís Enrique Lotte Aldinger		
				3	Work session 1 Exercise 1: Reciprocal diagrams in Rhino + GH basics	Instructors		
				Quick recap of Exercise 1			Lotte Aldinger	
		II	Week 2 (10/1)	1	Lecture 2 Algorithmic design & thinking	Dr. Juney Lee		
				2	Tutorial 2 Single node bridge in GH (form to force)	Lotte Aldinger		
				3				
			Week 3 (10/8)	1	Work session 2 Exercise 2: Single node bridge with GH	Lotte Aldinger + instructors		
				2				
		3						
		2D GS (procedural)	III	Week 4 (10/15)		Quick recap of Exercise 2	Lotte Aldinger	
					1	Lecture 3 Computational graphic statics	Dr. Juney Lee	
	2				Tutorial 3 Multi-node bridge in Grasshopper	Dr. Lluís Enrique		
	Week 5 (10/22)			3				
				1	Work session 3 Exercise 3: Multi-node bridge with GH	Dr. Lluís Enrique + instructors		
		2						
	Week 6 seminar week							
	trusses	IV	Week 7 (11/5)		Quick recap of Exercise 3	Dr. Lluís Enrique		
				1	Lecture 4 Algebraic graph statics (AGS)	Dr. Juney Lee		
				2	Tutorial 4 compas_aggs + IGS (Interactive Graphic Statics) plugin	Ricardo Avelino		
Week 8 (11/12)			3					
			1	Work session 4 Exercise 4: Truss problems with IGS	Ricardo Avelino + Instructors			
			2					
2. surface structures	shells	V	Week 9 (11/19)		Quick recap of Exercise 4	Ricardo Avelino		
				1	Lecture 5 Thrust Network Analysis (TNA)	Dr. Juney Lee		
				2	Tutorial 5 compas_tna + RV2 (RhinoVAULT 2) plugin	Dr. Juney Lee		
			Week 10 (11/26)	3				
				1	Work session 5 Exercise 5: Shell design exercises with RV2	Dr. Juney Lee + instructors		
				2				
		3. spatial structures	polyhedral structures	VI	Week 11 (12/3)		Quick recap of Exercise 5	Dr. Juney Lee
						1	Lecture 6 3D graphic statics (3GS)	Dr. Juney Lee
			Week 12 (12/10)		Quick recap of Exercise 6	Dr. Juney Lee		
				1	Lecture 7 Pt I. Outlook on computational graphic statics Pt II. Guest Lecture	Dr. Juney Lee TBD		
				2	Q/A	Discussions, feedback, evaluations, etc.		

Week 1 Friday, September 24th

Hour 1

9:45 – 10:30

- Welcome & introduction
- **Lecture 1: Introduction to graphic statics**
- Basics of graphic statics

10:30 – 10:45

- Mandatory 15 min break

Hour 2

10:45 – 11:30

- **Tutorial 1: Introduction to Rhino & Grasshopper**

11:30 – 11:45

- Mandatory 15 min break

Hour 3

11:45 – 12:30

- Introduction of exercise 1
- **Work session 1**

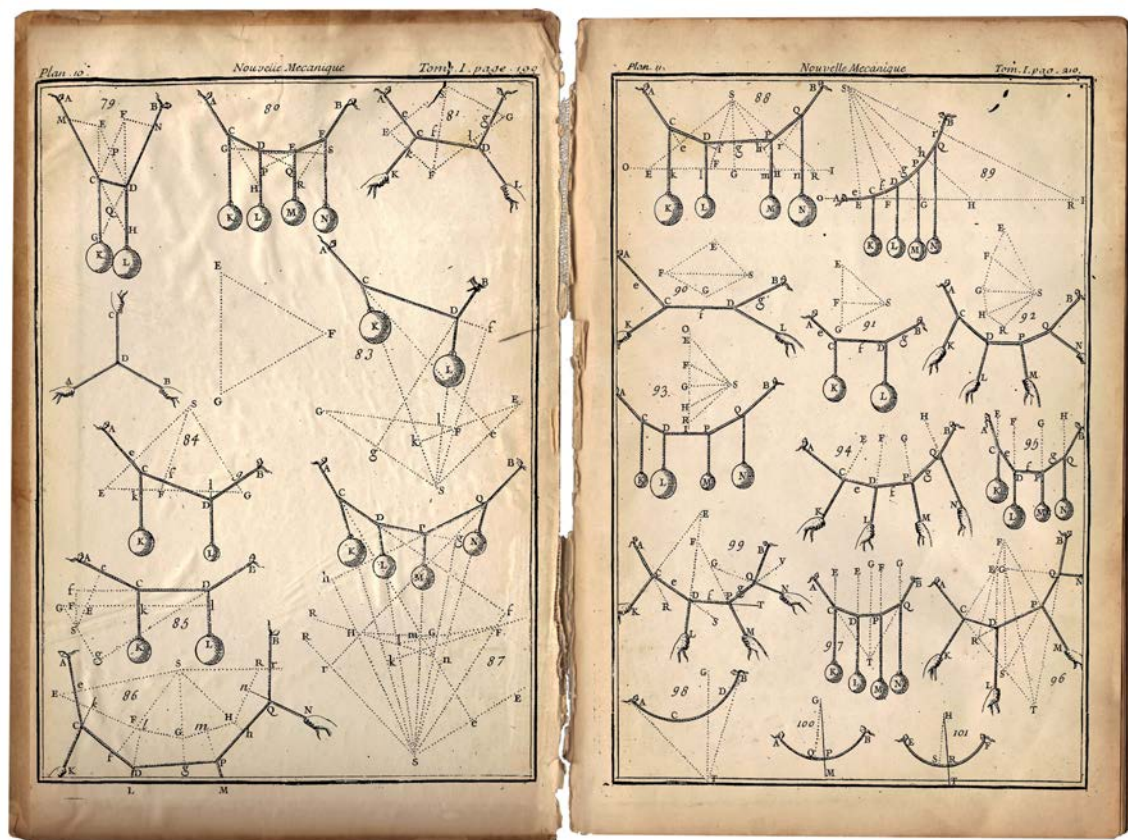
063-0605-00L : Computational Structural Design 1
Computational Graphic Statics

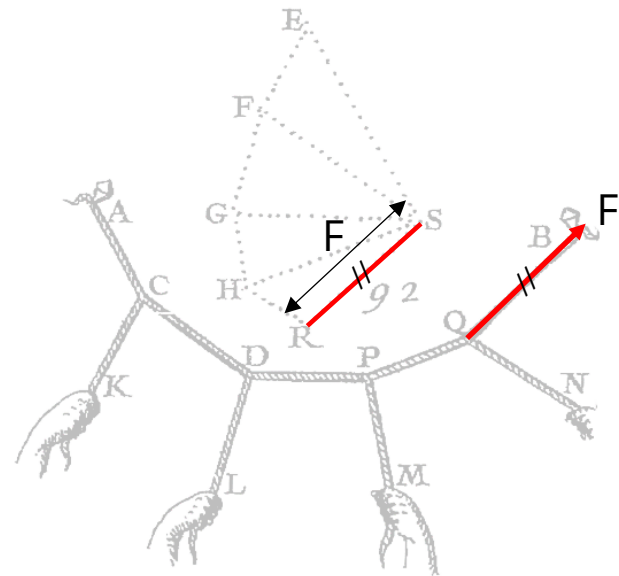
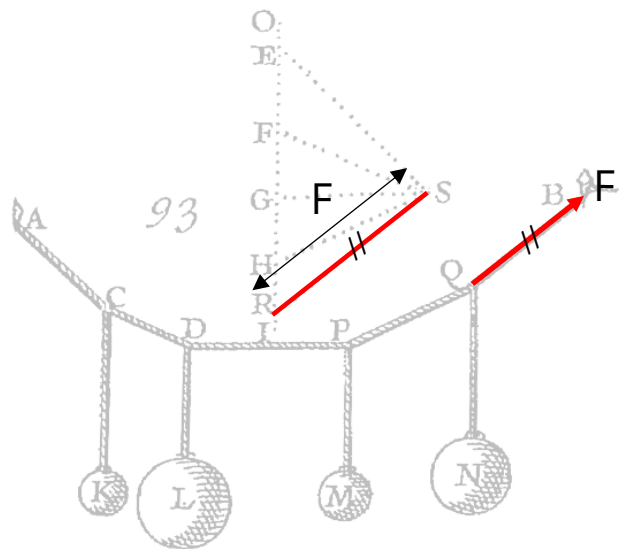
Lecture 1

Introduction to Graphic Statics

Friday, September 24th, 2021

Dr. Juney Lee
Dr. Lluís Enríque

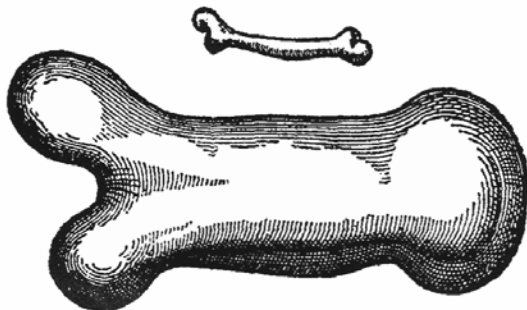




Stress



Material failure

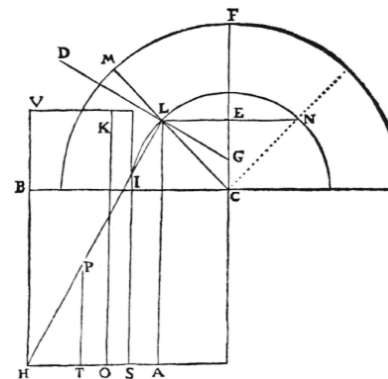


Galilei (1638)

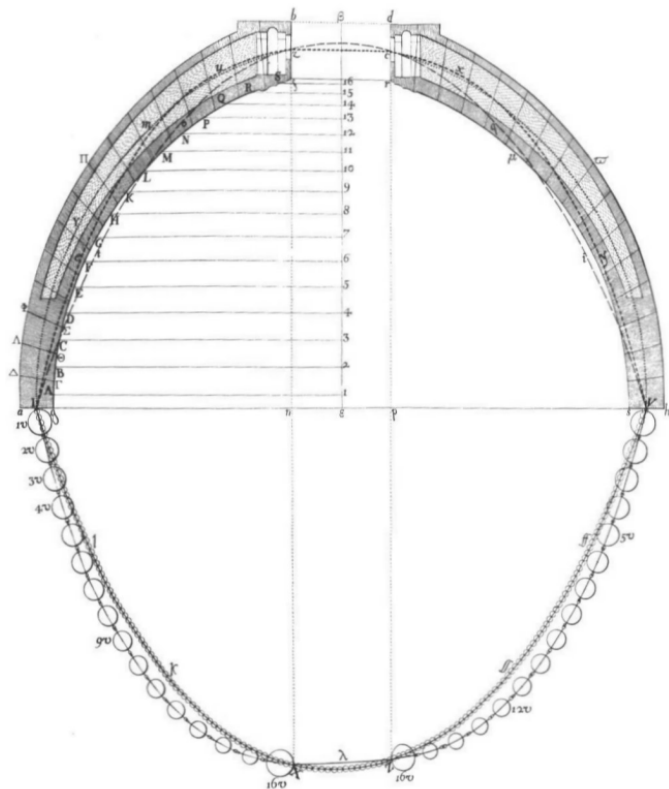
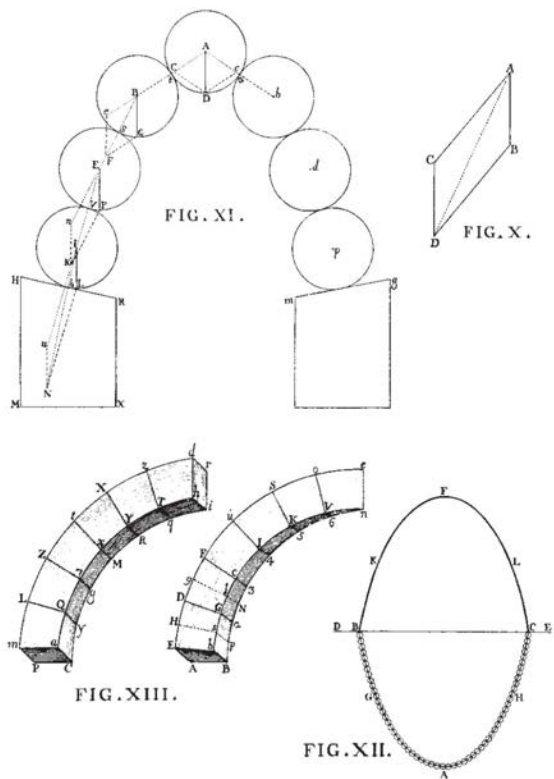
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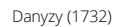


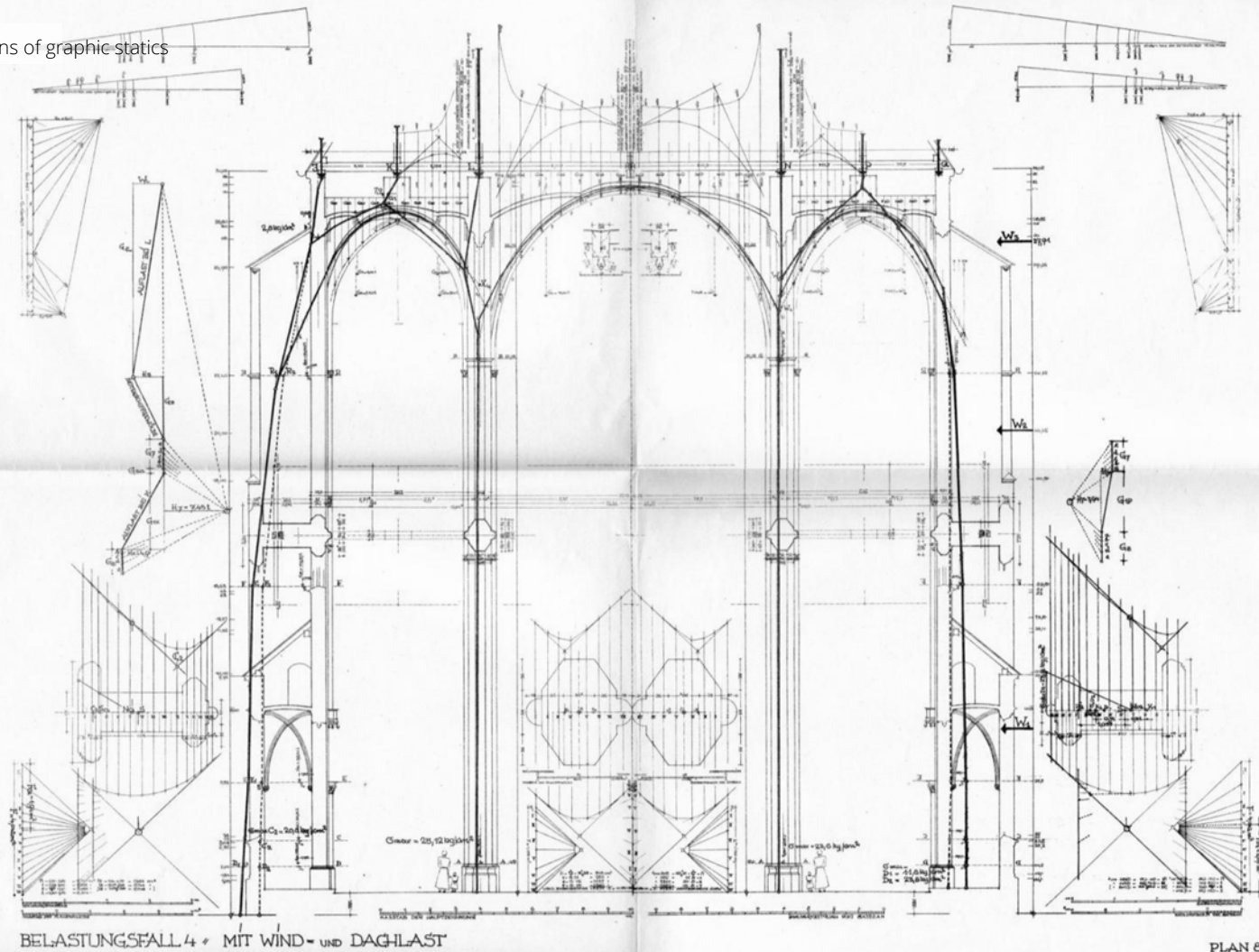
Geometry

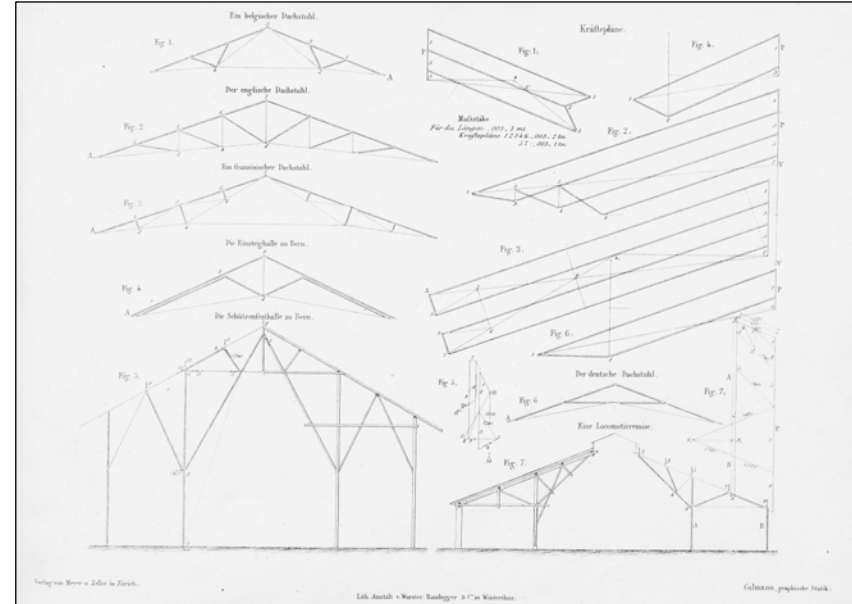
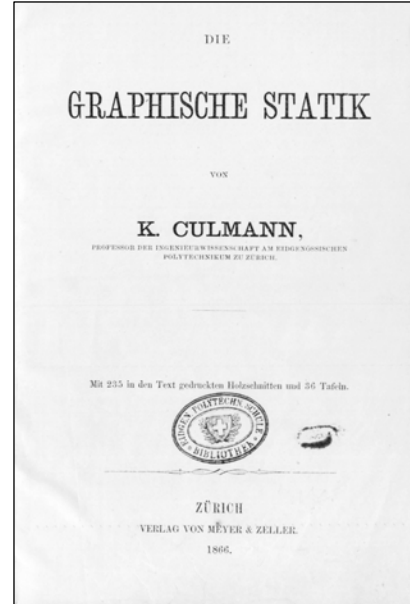


de Lahire (1695)









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the upper extremity of the segment 1 meets the upper extremity of the segment 2; and the straight line 18 through the point (16, 1), which is both the lower extremity of the segment 16 and that of the segment 1.



Fig. 8a.

Pass on to the joint (2, 17, 19, 20). Draw 19 through the point (17, 18), and 20 through the point (2, 3), the lower end of 2 and upper extremity of 3; and we obtain the polygon 2, 17, 19, 20, which is a rectangle.

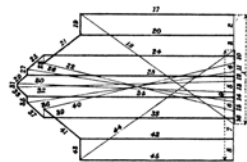


Fig. 8b.

Construct the polygon corresponding to the joint (16, 18, 19, 21, 22). For this purpose draw 21 through the point (19, 20), and 22 through the point (15, 16); we thus obtain a crossed pentagon. Continue to deal in the same manner with each of the points of application of the forces 8, 15, 4, 14, 13, 5, 12, 6, 11, 7, 10, 9, taken in succession.

Since the diagram *a*, which represents the skeleton of the structure and all the external forces, has for its axis of symmetry the vertical which passes through the centre of the figure, the diagram *b* has for its axis of symmetry the median horizontal line. For example, the triangle 9, 45, 44 is sym-

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metrical to the triangle 1, 17, 18; the rectangle 8, 45, 43, 42 to the rectangle 2, 17, 19, 20; and so on.

All the upper bars are in compression, and all the lower ones are in tension.

The diagonals and contra-diagonals are all in compression; finally two of the verticals 23, 39 are in tension, and all the rest in compression.

44. Figure 9a* represents one half of a locomotive shed. The external forces are the weights 1, 2, 3, 4, 5 applied at

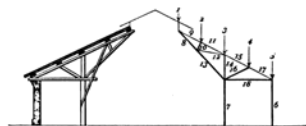


Fig. 9a.

the upper joints of the frame, and the reactions 6 and 7 of the wall and column. Again, all the external forces are parallel, and consequently the polygon of forces reduces, in diagram *b*, to one straight line. The force 8 (taken in the opposite sense to that in which it really acts) is equal to a certain part of the weight 5; by adding the difference to the other weights we get the magnitude of the force 7.

In the diagram *b* the direction of the lines 8 and 13 coincide; the first is a part of the second. Here then we have for the polygon corresponding to the joint (8, 10, 12, 13) one of those degenerate forms about which we spoke in Art. 33; the polygon is in fact a quadrilateral 8, 10, 12, 13, having three of its vertices (13, 8), (8, 10), (12, 13) in one straight line.

The polygon 5, 17, 18, 6, corresponding to the point where

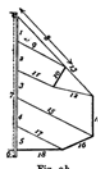


Fig. 9b.

* This example is taken from Pl. six. of the atlas of *Graphische Statik* of CULMANN, 1st edition. As previously stated, the two diagrams are not rigorously reciprocal.

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the roof is supported by the wall, presents an analogous degenerate form, since the vertices (6, 5), (upper point of the segment (6), (5, 17), and (18, 6) all lie in the same straight line.

The lower bars 8, 13, 18 are in compression, as well as the diagonals 10, 14, 16, the column 7 and the wall 6; while the upper pieces 9, 11, 15, 17 and the diagonal 12 are in tension.

45. Diagram *a* of Fig. 10 represents a truss at the upper joints of which are applied the oblique forces 1, 2, ..., 7, which

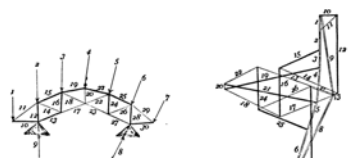


Fig. 10a.

may be considered as the resultants of the dead-loads and wind pressure; the forces 8, 9 represent the reactions at the supports.

The polygon of external forces is drawn in diagram *b* with double lines.

We construct successively the triangle 1, 10, 11, the quadrilateral 9, 10, 12, 13, the pentagon 2, 11, 12, 14, 15, the quadrilateral 13, 14, 16, 17, the crossed pentagon 3, 15, 16, 18, 19; the crossed quadrilateral 4, 19, 20, 21, the pentagon 17, 18, 20, 22, 23, and so on.

The upper bars 15, 19, 21, 25 are in compression, as well as the lower bars 10, 13, 30, and the verticals 12, 16, 24, 28; whilst all the remaining bars of the structure are in tension.

46. The diagram *a* of Fig. 11 represents a suspension bridge, loaded at each of its upper joints with weights 1, 2, ..., 9, and at each of its lower joints with weights 10, 11, 21, ..., 16; the weights are kept in equilibrium by the two

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oblique reactions 9, 17 at the two extreme points of the structure*.

The polygon of external forces has its first eight sides in succession along the same vertical straight line, and its seven



Fig. 11a.

last sides situated in another vertical straight line. The oblique forces 9 and 17 intersect, so that the polygon is a crossed one. We construct successively the polygons 1, 17, 19, 18; 16, 19, 20, 21; 2, 18, 20, 22, 23; 15, 21, 22, 24, 25; 3, 23, 24, 26, 27; and so on; most of which are crossed.

Diagram *b* shows that the upper bars are all in tension, and that the tension decreases from the ends towards the

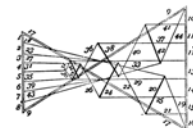


Fig. 11b.

middle of the structure; the bars of the lower boom are also all in tension, but in them the tension decreases from the middle towards the ends.

The extreme diagonals and contra-diagonals are in tension; in the portion situated to the left of the axis of symmetry, the diagonals or braces are alternately in tension and compression; similarly they are on the right but in the reverse order. Considering separately the ties and struts, we see that the internal

* This example is analogous to one of those studied by Maxwell in his memoir of 1870.

PLATE VII.

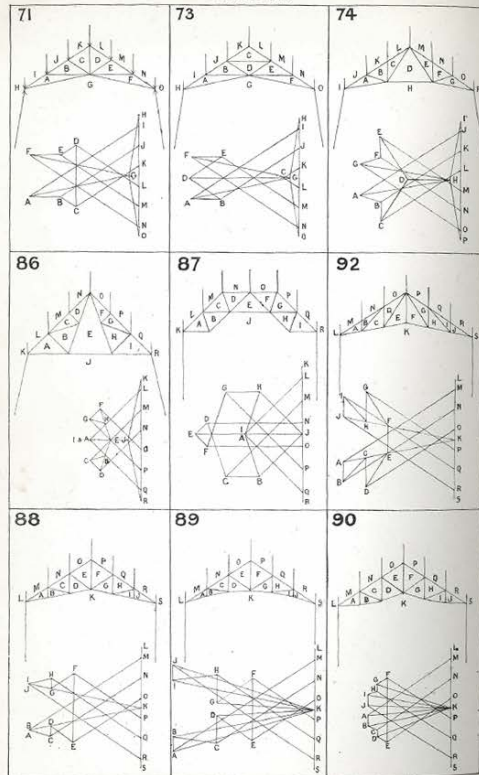
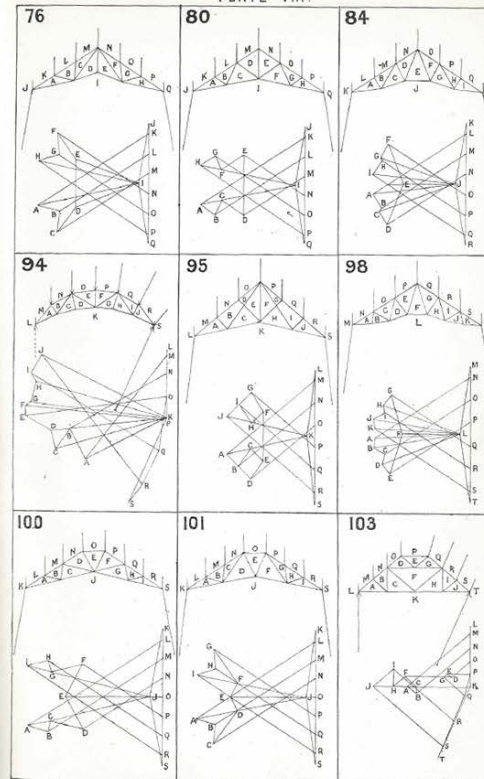
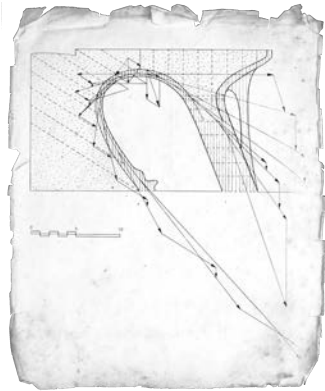


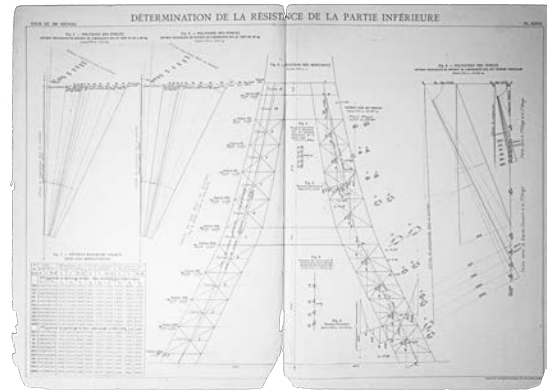
PLATE VIII.





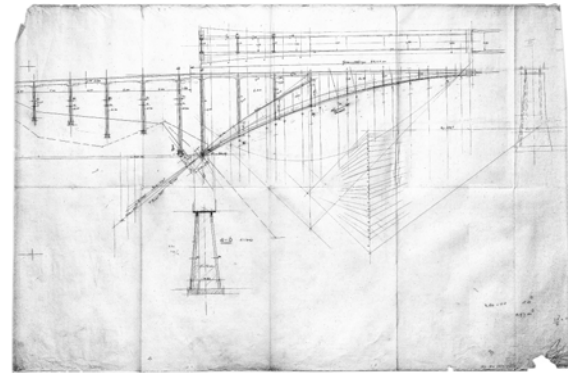
Park Güell | Gaudi (1914)

Rubio 1913, Tarrago 1991



Eiffel Tower | Gustave Eiffel (1889)

Wiki Commons



Salginatobel Bridge | Robert Maillart (1914)

Robert Maillart Archives



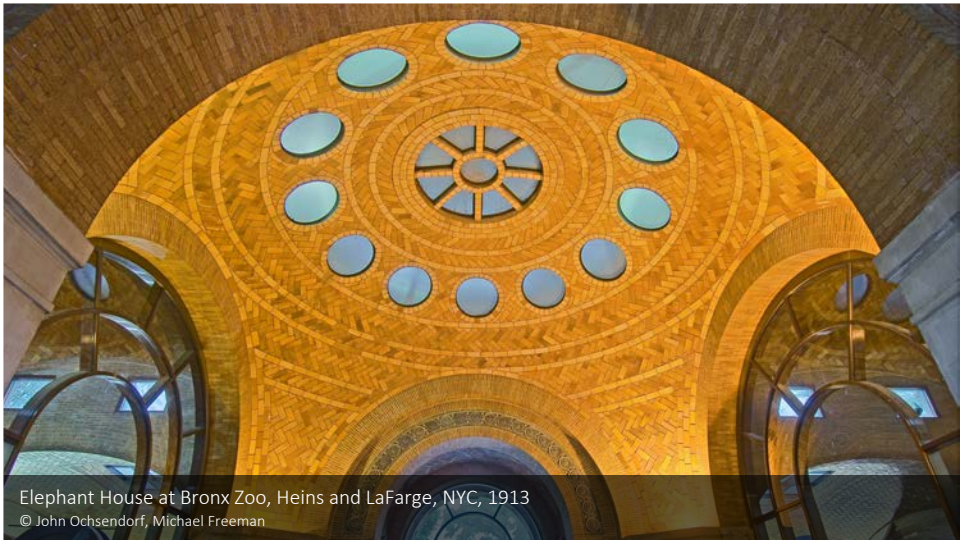
Oyster Bar, Grand Central Terminal, Warren and Wetmore, NYC, 1912-1913
© John Ochsendorf, Michael Freeman



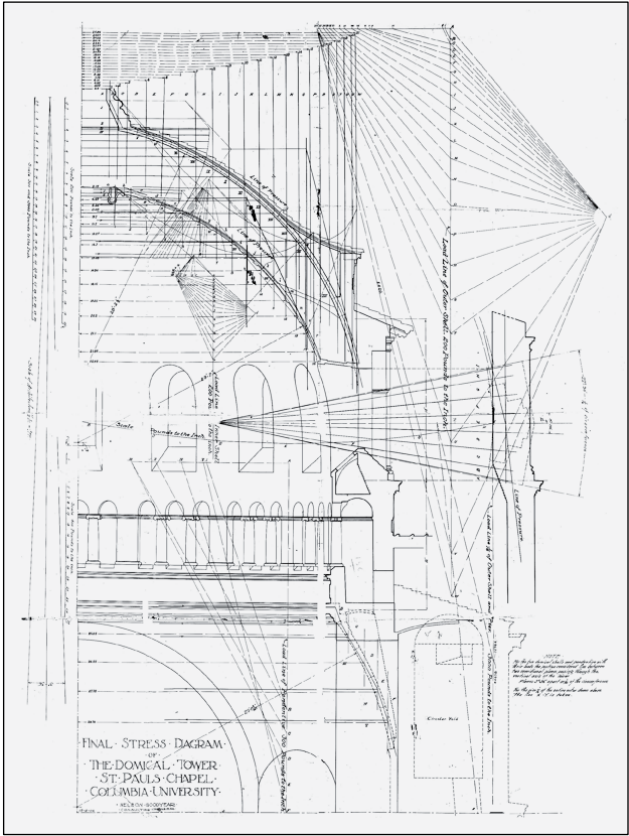
City Hall Subway Station, NYC, Heins and LaFarge, 1904-1913
© John Ochsendorf, Michael Freeman

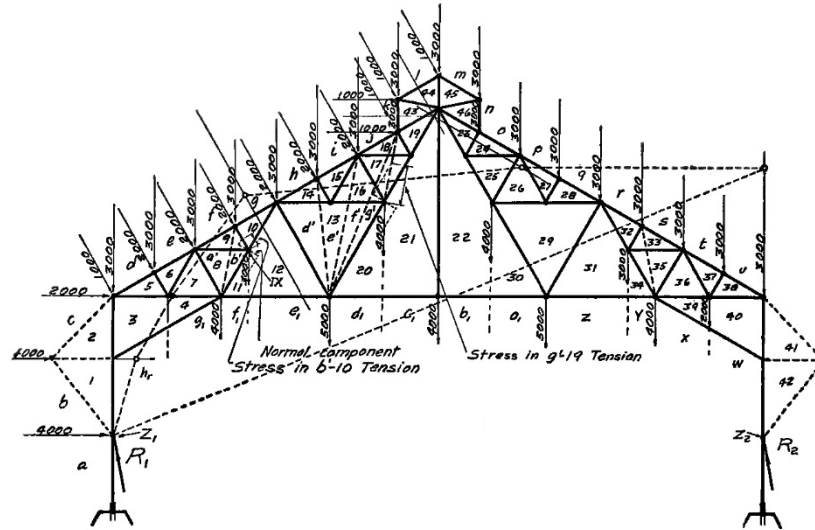


Queensborough Bridge Market, NYC, Palmer and Hornbostel, 1917-1913
© Michael Freeman

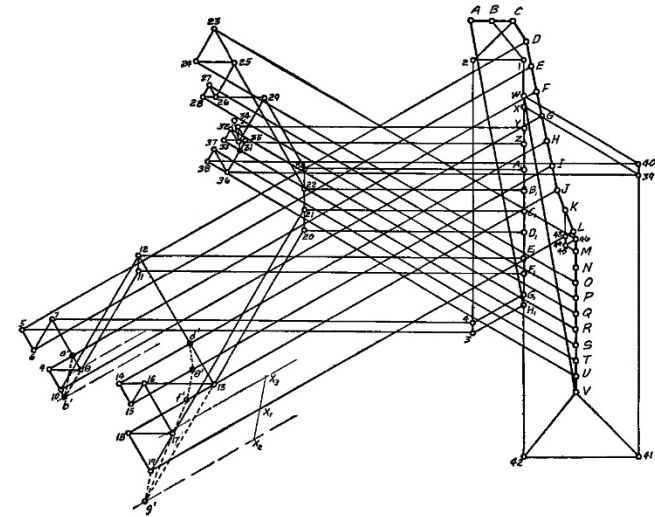


Elephant House at Bronx Zoo, Heins and LaFarge, NYC, 1913
© John Ochsendorf, Michael Freeman





Form Diagram



Force Diagram

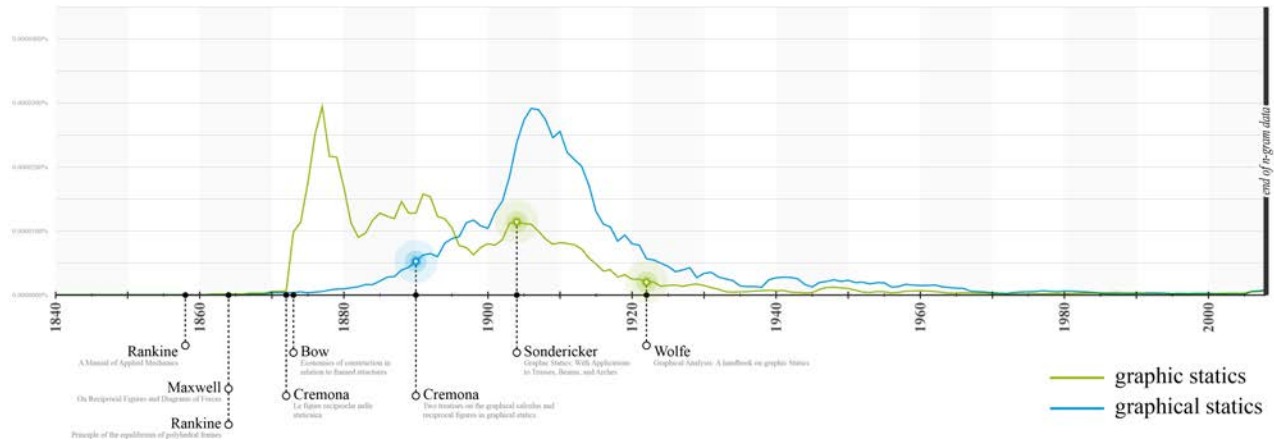
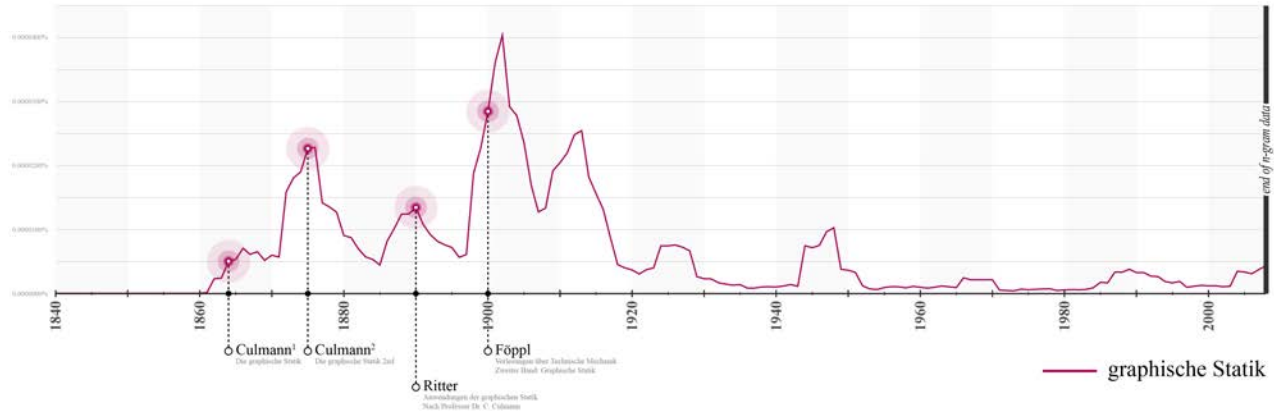
TOTAL LOAD ON
ONE HANGER

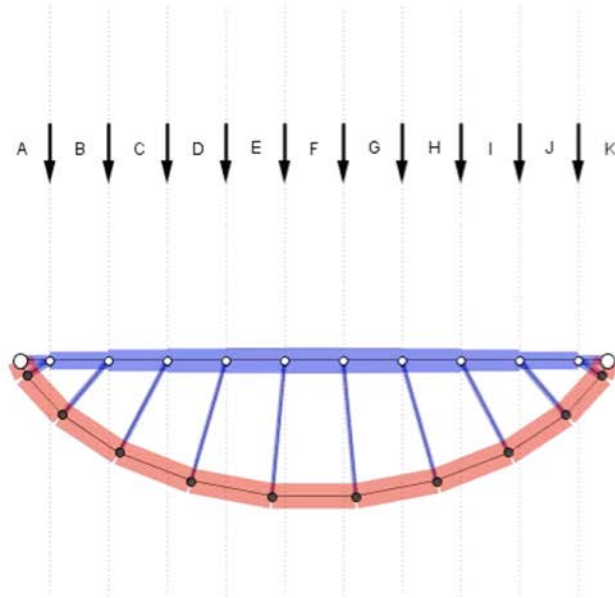
LOAD = 60 psf

$P = 60 \text{ lb} (400 \text{ ft})$

$P = 24,000 \text{ lb}$
 $= 24 \text{ K}$



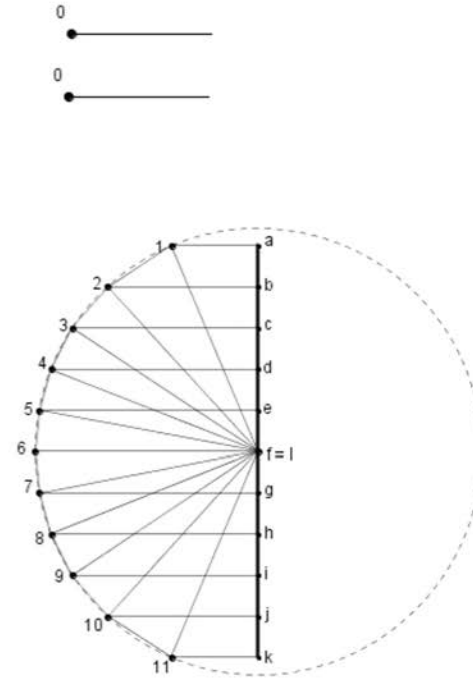




Form Diagram

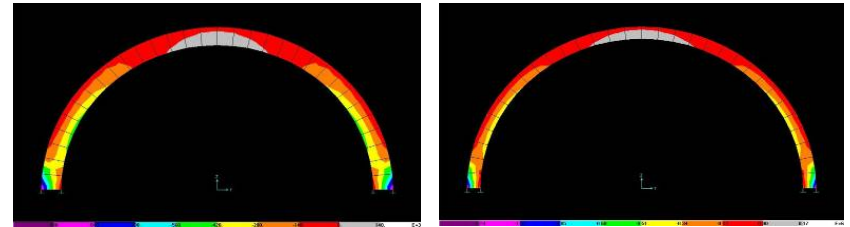
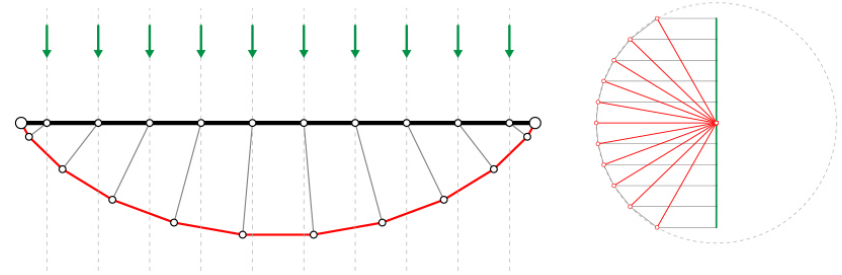
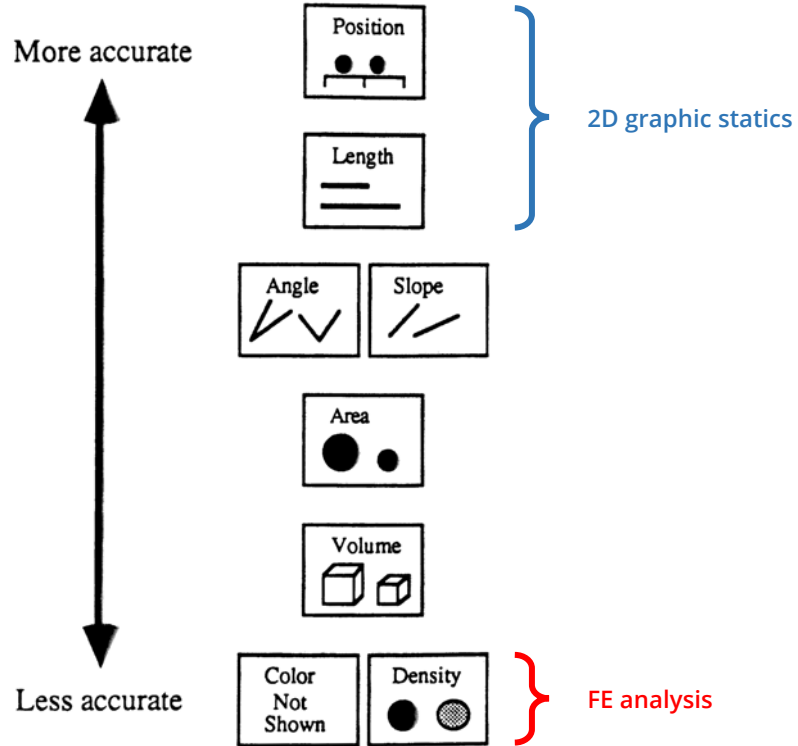
Geometry of the Truss Structure

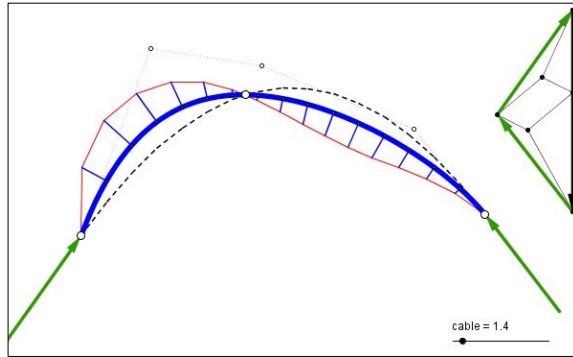
+



Force Diagram

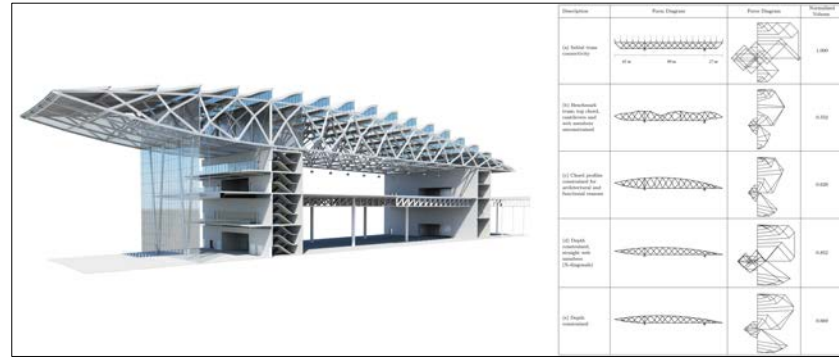
Equilibrium of Internal Forces





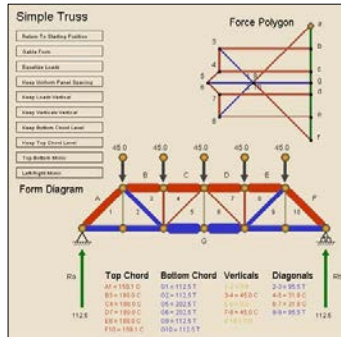
eQUILIBRIUM

Van Mele et al. (2009-)



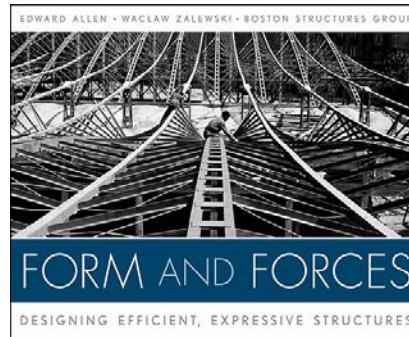
Structural Optimization using Graphic Statics

Beghini, et al. & SOM (2013)



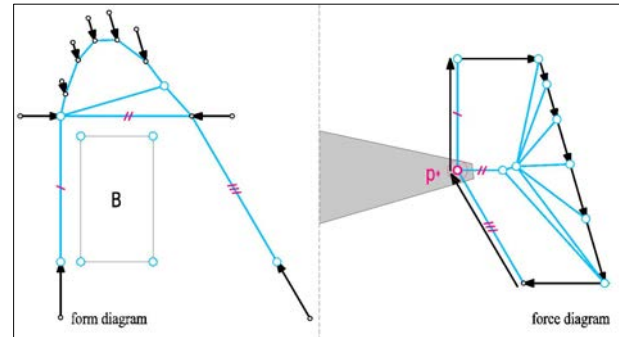
Active Statics

Greenwold & Allen (2003)



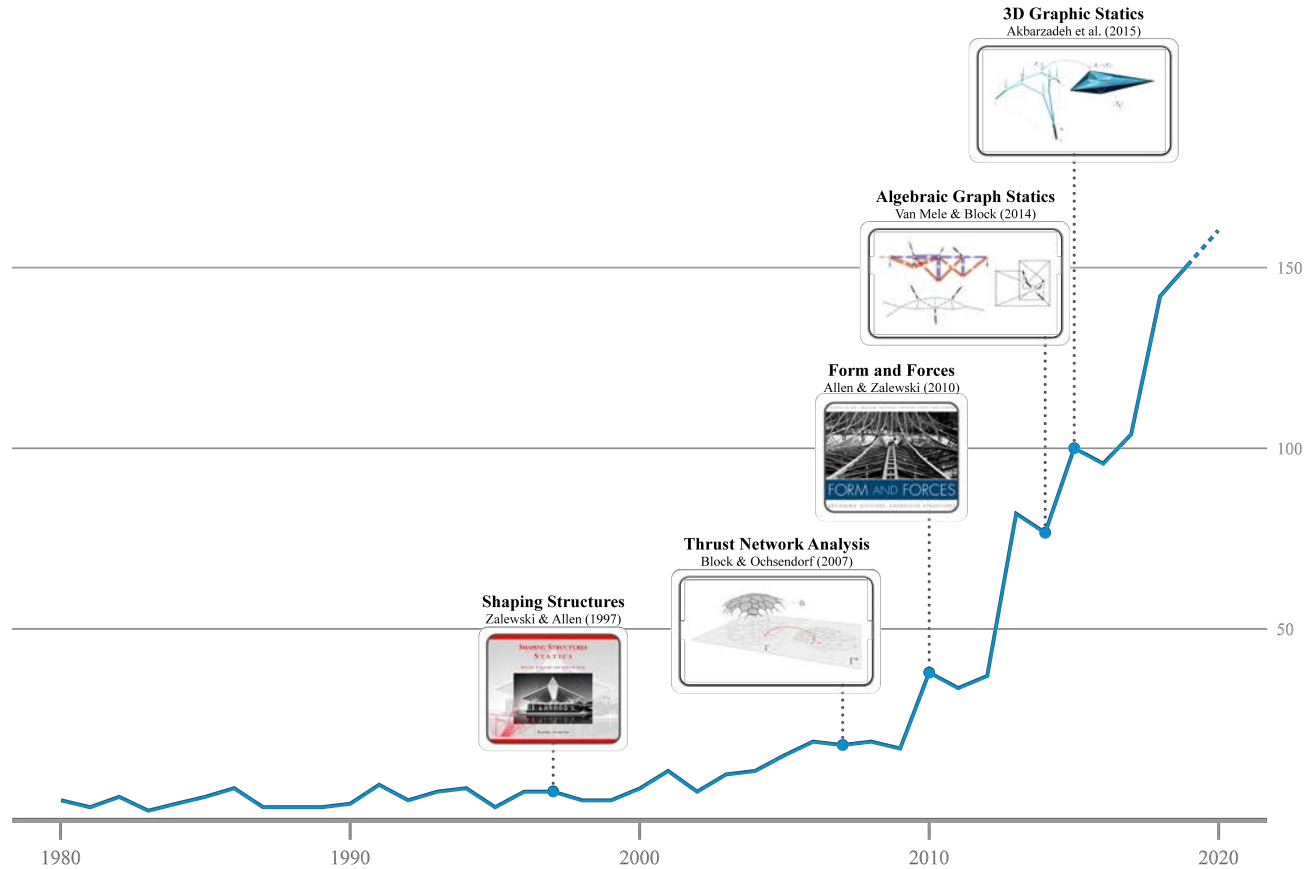
Form and Forces

Allen & Zalewski (2009)



Constraint-based Graphic Statics

Fivet & Zastavni (2013)











Karl Culmann

- Founder of graphic statics: “Die graphische Statik” (1864-1866)
- First Professor of Civil Engineering at the ETH Zurich (1855-1883)
- Director of the Polytechnic (1872-1875)



Karl Wilhelm Ritter (ETHZ diploma, 1868)

- Professor of graphic statics and bridge construction (appointed in 1882)
- Director of the Polytechnic (1887-1891)
- Extended applications of Culmann's graphic statics:
“Anwendungen der graphischen Statik nach Professor Dr. Culmann” (1888-1906)



Robert Maillart
(ETH diploma, 1894)



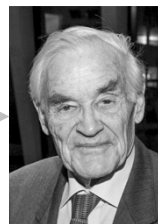
Othmar Ammann
(ETH diploma, 1902)



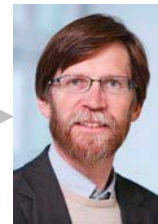
Pierre Lardy
(ETH diploma, 1928)



Heinz Isler
(ETH diploma, 1950)



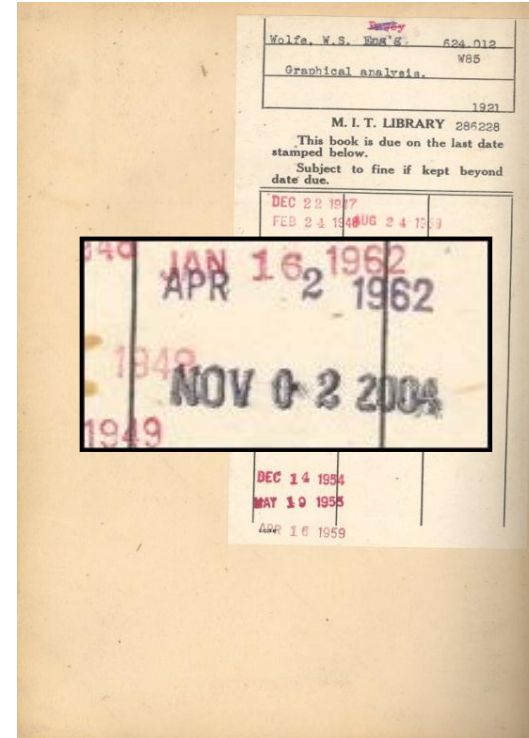
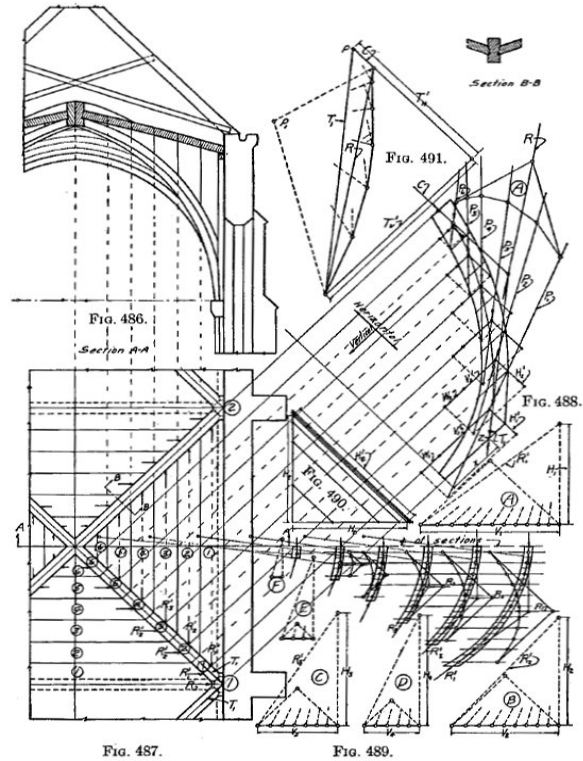
Christian Menn
(ETH diploma, 1950)

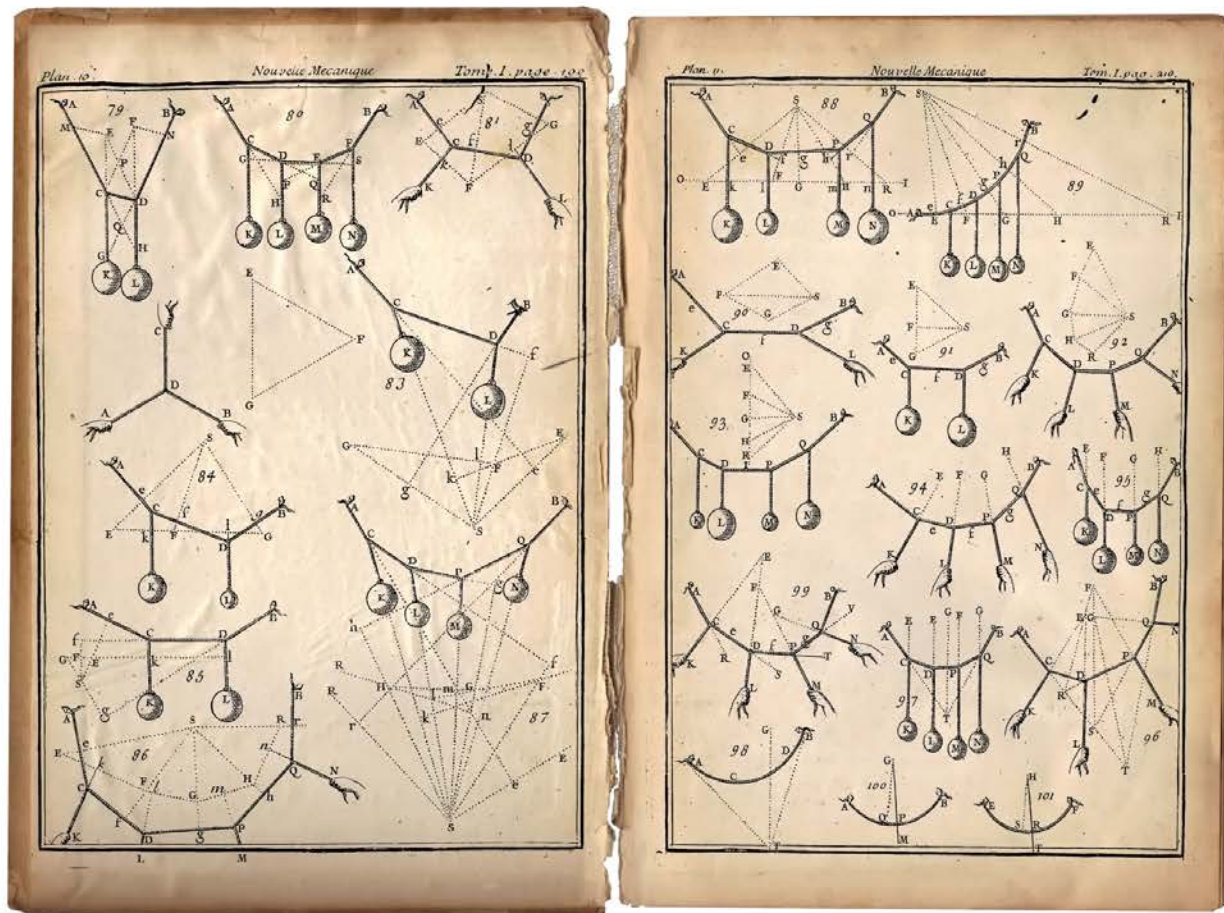


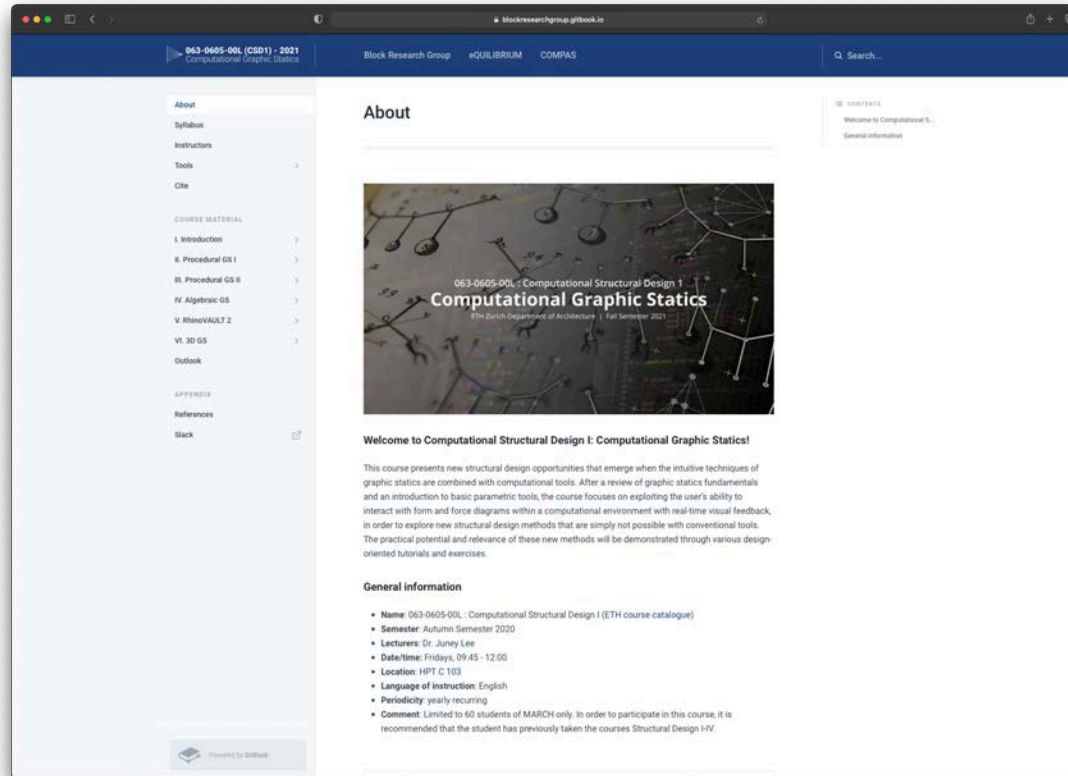
Joseph Schwartz
(ETH professor, 2008-)

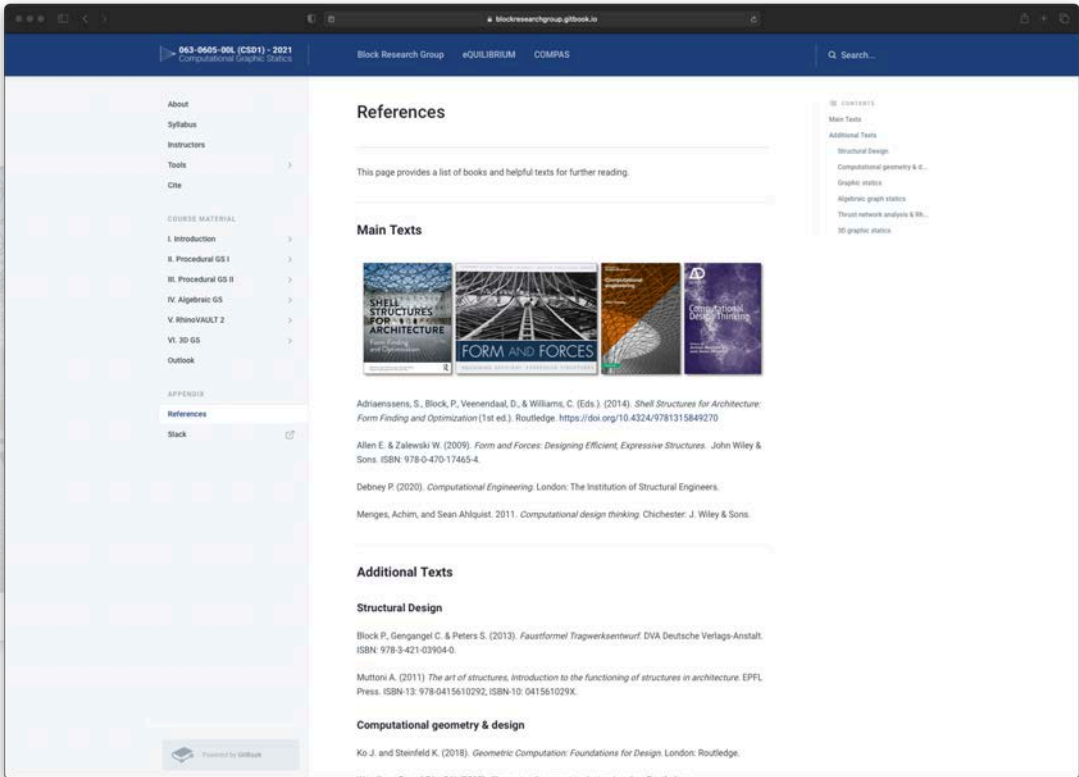


Philippe Block
(ETH professor, 2009-)









The screenshot displays the course GitBook website. The main content area is titled "References" and provides a list of books and helpful texts for further reading. The page is organized into two main sections: "Main Texts" and "Additional Texts".

Main Texts

- Adriaenssens, S., Block, P., Veenendaal, D., & Williams, C. (Eds.). (2014). *Shell Structures for Architecture: Form Finding and Optimization* (1st ed.). Routledge. <https://doi.org/10.4324/9781315849270>
- Allen E. & Zalewski W. (2009). *Form and Forces: Designing Efficient, Expressive Structures*. John Wiley & Sons. ISBN: 978-0-470-17465-4.
- Detmey P. (2020). *Computational Engineering*. London: The Institution of Structural Engineers.
- Menges, Achim, and Sean Ahlquist. 2011. *Computational design thinking*. Chichester: J. Wiley & Sons.

Additional Texts

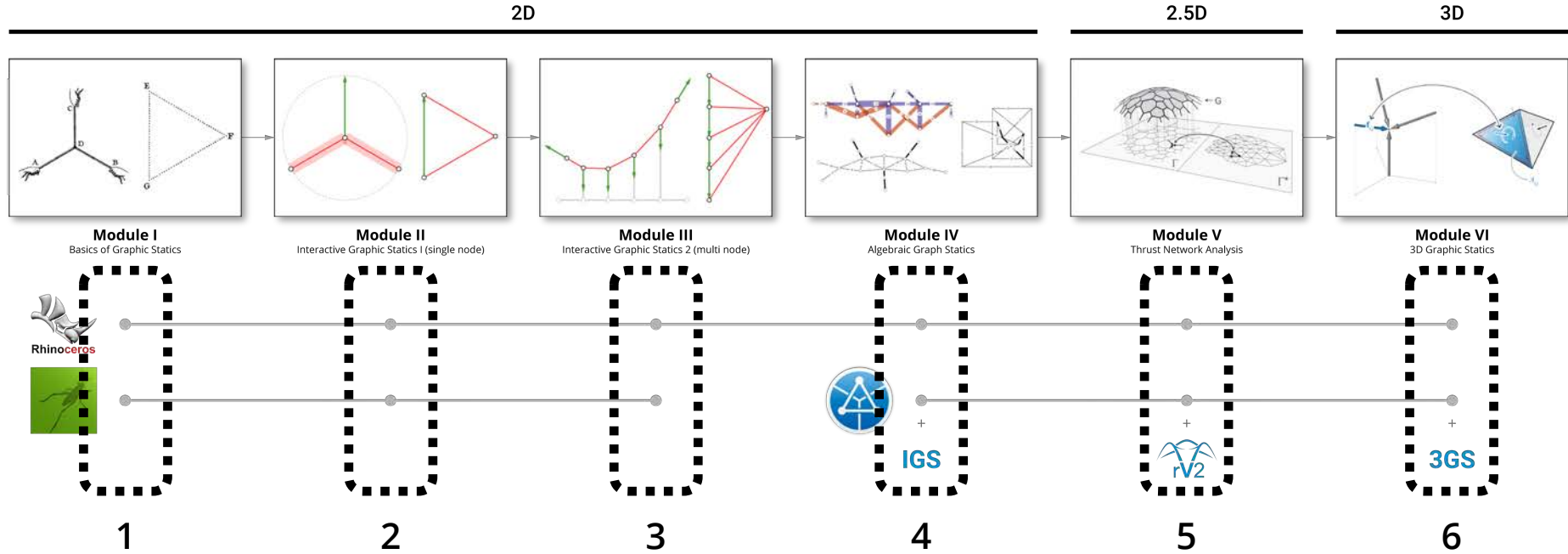
Structural Design

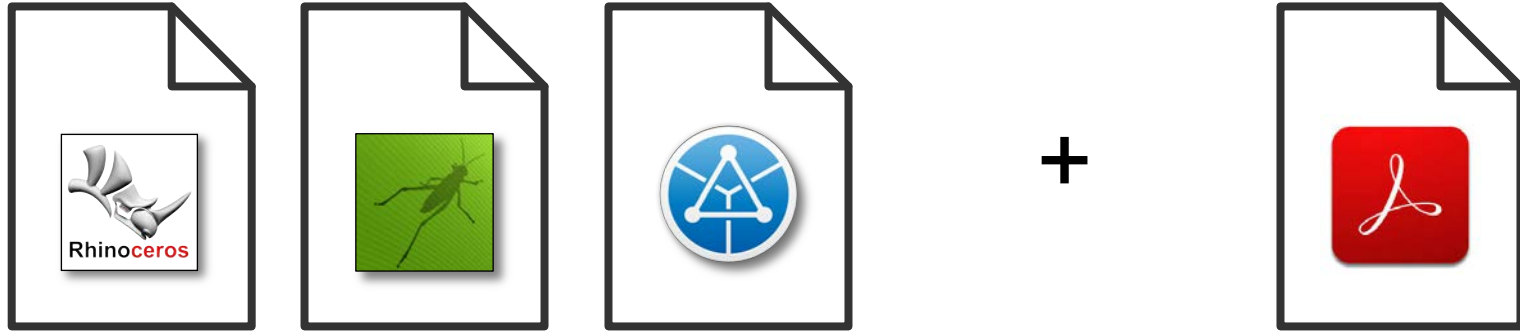
- Block P., Gengnagel C. & Peters S. (2013). *Faustformel Tragwerksentwurf*. DVA Deutsche Verlags Anstalt. ISBN: 978-3-421-03904-0.
- Muttoni A. (2011) *The art of structures, introduction to the functioning of structures in architecture*. EPFL Press. ISBN-13: 978-0415610292, ISBN-10: 041561029X.

Computational geometry & design

- Ko J. and Steinfield K. (2018). *Geometric Computation: Foundations for Design*. London: Routledge.

The left sidebar shows the course material structure, including sections like "About", "Syllabus", "Instructors", "Tools", "Cite", "COURSE MATERIAL", "I. Introduction", "II. Procedural GS I", "III. Procedural GS II", "IV. Algebraic GS", "V. Rhinoceros 2", "VI. 3D GS", "Outlook", "APPENDIX", "References", and "Stack". The right sidebar shows the table of contents, including sections like "CONTENTS", "Main Texts", "Additional Texts", "Structural Design", "Computational geometry & design", "Algebraic graph statistics", "Thrust network analysis & Rh.", and "3D graphic statistics".





Skill

Understanding

CSD1_FS2020

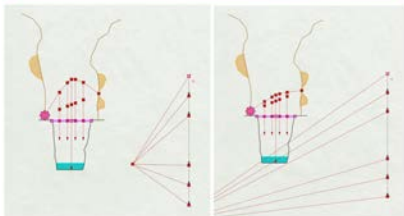
10/27/20 9:33 PM

3. The basic idea is to construct the limitations and let rhino show the warning by displaying a symbol at the footage if either of the 3 conditions is not satisfied.

1) To constrain the inclination on the left side: First, input the limit angles X, Y as number sliders and then use the "Rotate" component to translate them into vectors. Starting from the point A on force diagram, use "Line SDL" to draw two constrain lines based on those vectors and get two intersection points with the vertical line (let's name them as point "Bc" and point "By" in the force diagram. Deconstruct the two points and the top point which represents the first force line. By calculating those Y-coordinates (substrate them in pairs and then multiply the results and then to see if it positive or negative) one could easily inspect if the top point lies in between the two points "Bc" and "By".

2) To examine if the horizontal components at the left footage: Input the Z value and compare it with the distance of point A to the vertical line. (Deconstruct the point A and calculate the difference between the X-components of point A and the vertical line)

3) To exclude the tension situation: one could simply limit the allowable value of inputted X and Y value: all must bigger than "0".



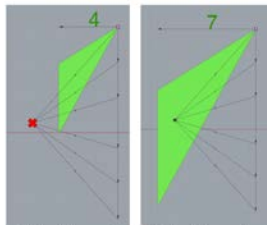
3

CSD1_HS2020

September 2, 2020 10:16 AM

3. The maximum horizontal force is defined by a parallel line to the load line from the force diagram. This line moves to the left, according to the maximal load set in the "max Horizontal support" component. The angles, in which the support can be, can be controlled with the two other components (angle 1 and angle 2). The move component is defined by the motion in "Unit x" (because the line needs to be moved along the x axis) and a reverse component (so the vector turns negative). Angle 1 and 2 are defined by a rotation. Angle 1 is the rotation of the direction of the load-line. Angle 2 is the rotation from angle 1. The direction and the size of the support forces are in this case A and B.

Is the support force A inside the green surface, so the direction of the support force is ok, for the defined angles of Angle 1 and Angle 2. The pole inside the green surface, the direction and the size of the maximal horizontal component are ok for the set parameters.



Direction of the support force inside the set parameters, but the set horizontal component is too small (or the geometry of the bridge too shallow)

Direction of the support force inside the set parameters. horizontal component also inside the set parameters
→ no cross at the pole

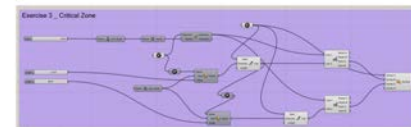
2

CSD1_FS2020

10/29/20 11:42 PM

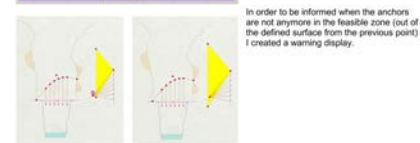
The result of the division can not be a round number because in the way I proceeded it will always be an odd number, that's why I used the "Round" component. I then linked the result with index output of the List item which is also connected to the Panel component of the force magnitudes. I finally connected the first List to the existing Amplitude component. That way I managed to add the force in the middle of the bridge.

Part 3 - Modify the parametric model to provide solutions only in the feasible design space.



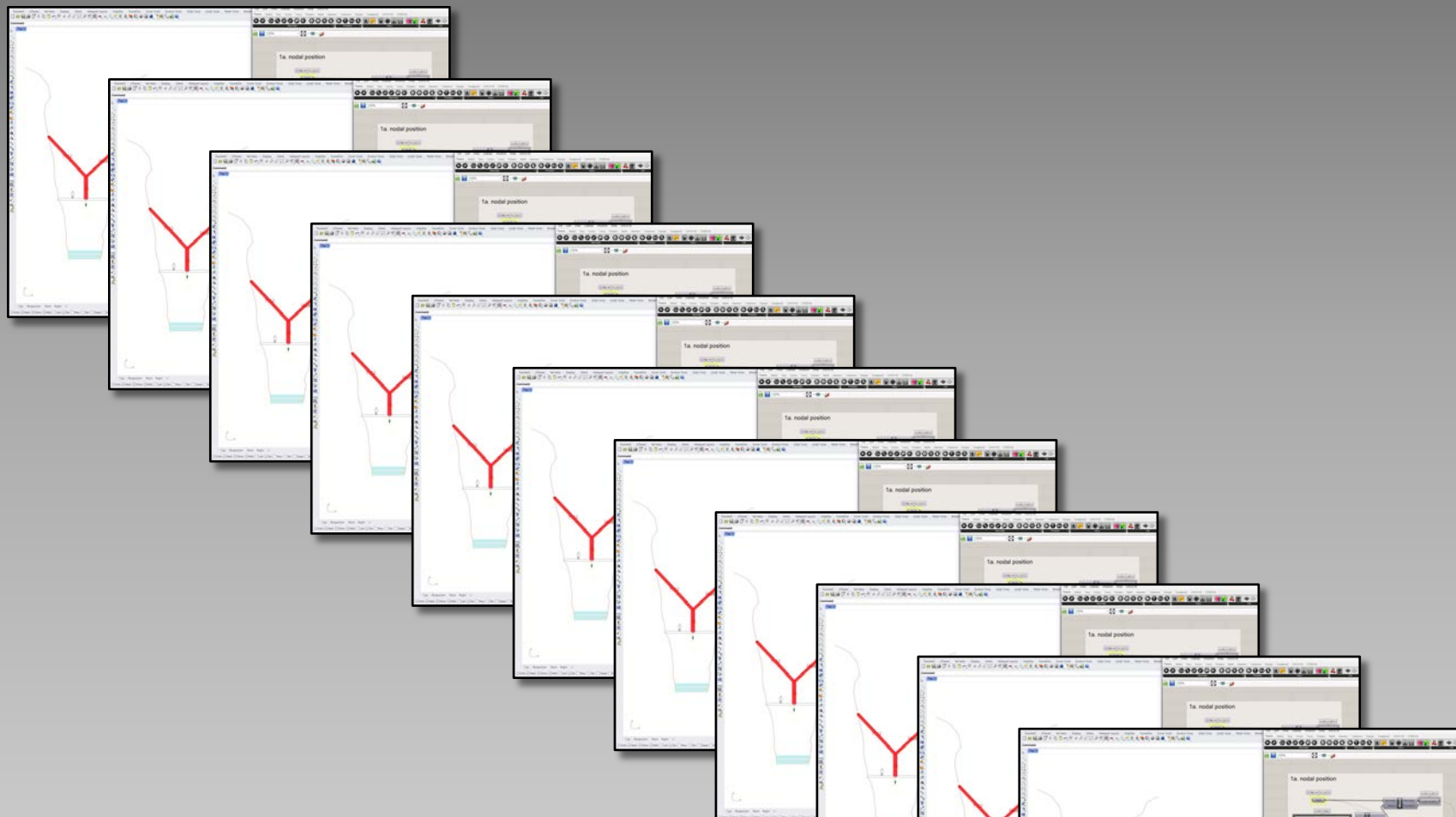
To create the asked limitation we start by adding the Hmax, angle a and angle b values with a number slider. The Hmax number slider connected to the Move component will define the intensity of the movement of the geometry according to the chosen Hmax value. Same for the angles a and b, which are connected to the Rotate components. The two angles linked to the Z unit Axis define the rotation of the vector line.

With the Line SDL component I created the vectors starting from the initial point in the force diagram which followed the defined direction of the previous Rotate component. The Line / Line component allowed me to create the intersecting points between the two vectors of the angles and the line defined by Hmax. I finally used the 4 point Surface component to link the line defined by the Hmax and the two vector lines of the angles in order to get the feasible region.



In order to be informed when the anchors are not anymore in the feasible zone (out of the defined surface from the previous point) I created a warning display.

2



The screenshot displays the 'Exercise 1' page on the Block Research Group website. The page is titled 'Exercise 1' and contains instructions for completing tasks and submitting a zipped folder. The sidebar on the left lists various course materials, with 'Exercise 1' highlighted. The main content area includes a list of tasks and a download link for 'CSD1_exercise-1.3dm'. The bottom of the page features a preview of the eEQUILIBRIUM interface, which is also highlighted by a red arrow.

063-0605-00L (CSD1) - 2021
Computational Graphic Statics

Block Research Group eEQUILIBRIUM COMPAS

Search...

About
Syllabus
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Cite

COURSE MATERIAL

I. Introduction
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Tutorial 1
Exercise 1

II. Procedural GS I
III. Procedural GS II
IV. Algebraic GS
V. RhinovaULT 2
VI. 3D GS
Outlook

APPENDIX
References
Stack

Exercise 1

Complete the tasks below, and submit a zipped folder that includes the following three files by 9:00am on Friday, October 1st.

1. Part 1 - Rhino file with completed tasks
2. Part 2 - completed Grasshopper definition
3. Summarising PDF

Please follow the file naming convention as shown in the [Syllabus](#).

[Submit Exercise 1 here.](#)

Part 1 - Graphic statics in Rhino

Complete the following 4 tasks using this Rhino file.

[CSD1_exercise-1.3dm](#) CSD1_exercise-1.3dm - 404KB

Task 0 - Review eEQUILIBRIUM drawings

eEQUILIBRIUM, an interactive environment for graphic statics-based structural design, provides examples of pre-constructed graphic statics drawings. These drawings are interactive and have various features that can be used to learn various fundamental principles of graphic statics.

For the first task of this exercise, simply check out the first two rows of the "drawings" page on the eEQUILIBRIUM platform, and learn the principles and construction techniques demonstrated in each drawing.

Powered by Gitbook

Block Research Group sent you this request.

CSD1_2021_exercise-1

Upload by Oct 1, 2021 at 9:00 AM GMT+2

Add files

or drag stuff here

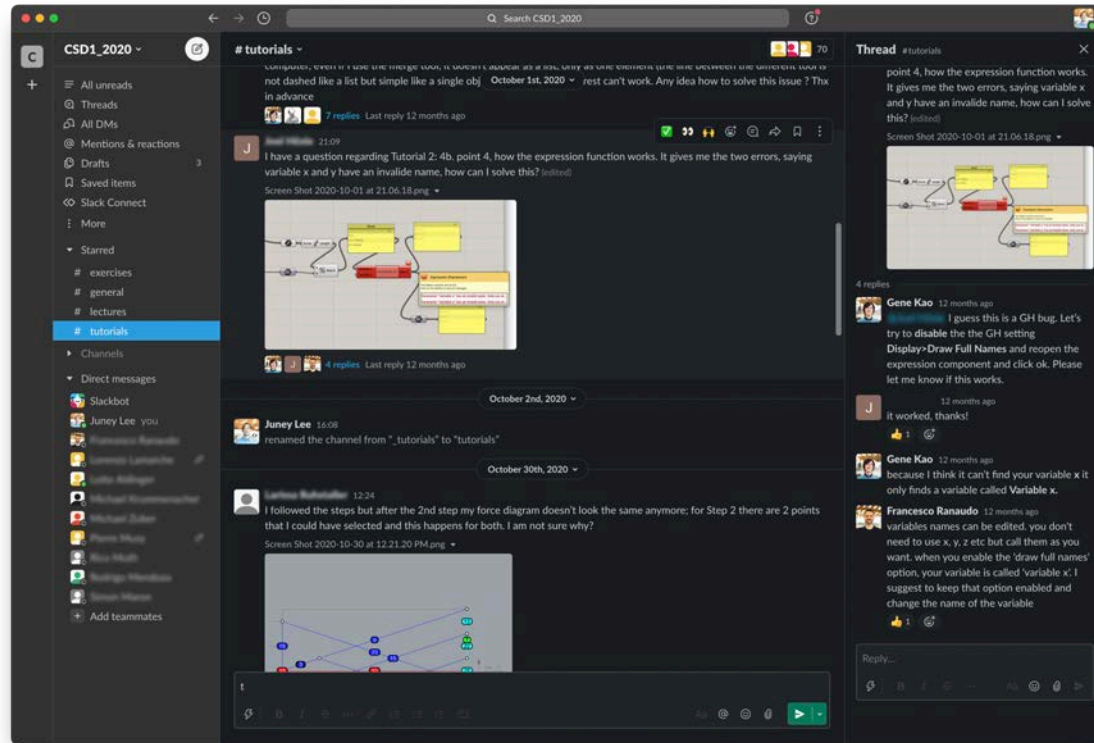
Your files will be uploaded securely to Block Research Group's Dropbox account. More about file requests and our privacy policy.

Cookie Preferences | English (United States)

File naming convention

exercise-number_item-number_firstname-lastname.extension

- exercise-1_jane_smith.zip
 - exercise-1_task-1_jane-smith.3dm
 - exercise-1_task-1_jane-smith.gh
 - exercise-1_task-2_jane-smith.3dm
 - exercise-1_task-2_jane-smith.agx
 - ...
 - exercise-1_jane-smith.pdf



Course website → Link provided on the course GitBook

063-0605-00L : Computational Structural Design 1
Computational Graphic Statics

<http://block.arch.ethz.ch>

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